DDA Scattering Pipeline — Project Manual

Single scattering properties of randomly oriented hydrometeors (ADDA → ARTS SSDB)

2026-11-10

Contents

How to use this manual . . . 3

Part 0 — Orientation **6**
0.1 Repository map . . . 6

Part 1 — Theory **9**
1.1 The Discrete Dipole Approximation (DDA) . . . 9
1.2 Particle shape files and the resolution scheme . . . 10
1.3 Dielectric function of ice (Matzler 2006) . . . 12
1.4 From ADDA output to Mueller and extinction matrices . . . 13
1.5 Orientation averaging — the two strategies . . . 15
1.6 Spherical-harmonic expansion over the scattering angles . . . 17
1.7 Mirror-symmetry exploitation (Phase 4) . . . 18
1.8 What the SSDB export contains . . . 20
1.9 Analytic diffraction-peak model + residual storage (Phase 9A — scanned, not yet in the pipeline) . . . 20

Part 2 — Architecture & data flow **29**
2.1 ARO call graph (production path) . . . 29
2.2 TRO call graph . . . 29
2.3 Artifact data flow (what lives where, what is transient) . . . 29
2.4 Module status table . . . 29

Part 3 — Operations & runbooks **35**
3.1 The config contract . . . 35
3.2 ARO job flowcharts . . . 36
3.3 Post-processing: folder → SSDB NetCDF . . . 36
3.4 TRO production run . . . 36
3.5 Local (macOS) runbook . . . 43
3.6 HPC runbook (Mistral / Thunder) . . . 44

Part 4 — Validation & quality gates **45**
4.1 Golden baselines (`baselines/`) . . . 45
4.2 Equivalence suites (what was proven, and how) . . . 45
4.3 Where the accuracy lives (error budget recap) . . . 46
4.4 Production backlog (as of Phase 5) . . . 47

Appendix A — ADDA command-line tokens used **48**

Appendix B — the .dat input files **49**

Appendix C — SSDB NetCDF layout **50**

Appendix D — Gotcha catalog **51**

Appendix E — Baseline simulation results **54**
E.1 Headline numbers (frozen 1.2 outputs) . . . 54
E.2 Figures . . . 55
E.3 Reproduce . . . 57

References **58**

How to use this manual

This is the single reference document for maintaining and operating the DDA automation pipeline in this repository. It is written for the **future project maintainer**: enough physics theory to understand *what* is computed and *why* each knob matters, and enough operational detail (verified against the actual source, not from memory) to run, extend and debug the pipeline safely.

- Flowcharts / call graphs are **Mermaid** diagrams — they render on GitHub, in VS Code (Markdown preview), and in the generated HTML (`docs/DDA-manual.html`). In the PDF they become dedicated full-figure pages (auto-rotated sideways when that prints larger).
- All quantitative plots are **real data** (measured benchmarks, real shape metadata, the actual Matzler-2006 implementation). They are regenerated by `make_figures.py`[1]: `mamba activate DDA && python docs/make_figures.py` → `docs/figures/*.png`. If you change the backlog or run new benchmarks, re-run it and commit the figures. (The Appendix-E baseline plots are re-rendered by that script *delegating* to `baselines/plot_baseline_figs.py`, so the manual can never drift from the reference baseline report.)
- Every figure has a **bold *Fig. N* caption line directly below it** stating the one thing it proves and how to read it. If you only skim, read the figure captions.
- The **PDF** (`docs/DDA-manual.pdf`) is typeset with LaTeX: `python docs/build_latex.py` (pandoc → xelatex via latexmk; needs TeX Live + the `DDA` conda env with `pypandoc_binary` and a Chrome for the mermaid vector rendering). The browsable **HTML** version is built with `python docs/build_html.py` (add `--pdf` if a headless Chrome/Chromium is installed). Mermaid diagrams in the HTML render via mermaid.js from a CDN — same requirement as GitHub. The Markdown source `docs/DDA-manual.md` is the single source of truth for both — never hand-edit the `.tex`.

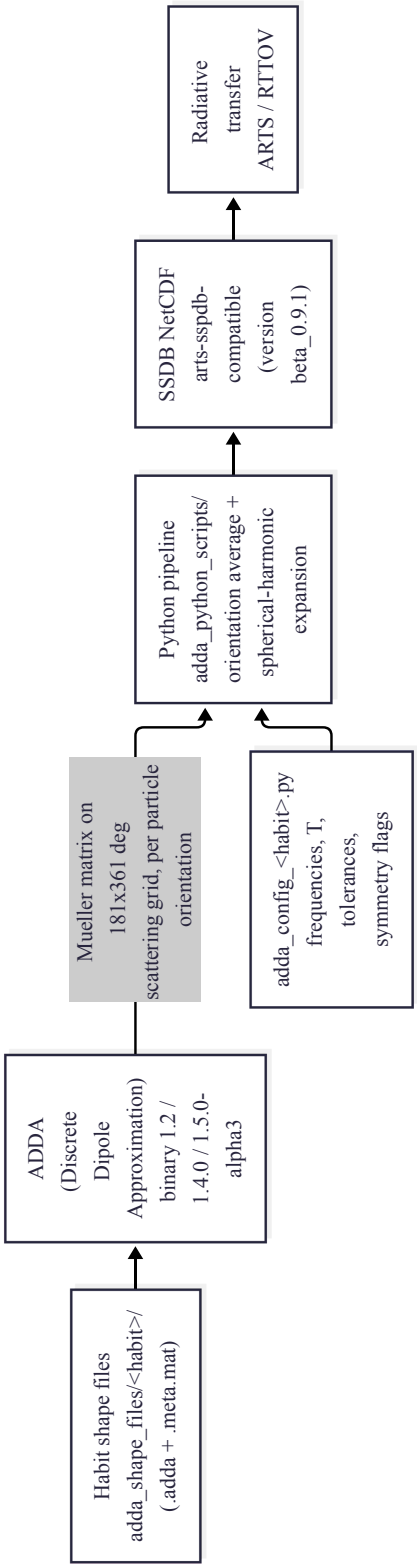
Pick your entry point

You want to…	Read
run a job today (local smoke test)	§0.3 status → §3.5 local runbook → §3.2 flowcharts
submit the production backlog on an HPC	§3.6 runbook → §4.4 job/cost table → Appendix A CLI tokens
understand *what* is computed and why a knob matters	Part 1 (theory), in order §§1.1→1.6
debug a run / extend the pipeline	§2 call graphs → Appendix D gotchas → `plans/DDA-modernization-report.md`
validate a change before merging	Part 4 gates + `baselines/` (§4.1)

The one-minute picture

The pipeline answers one question: *for a given particle habit, size, frequency and temperature, what are its single-scattering properties averaged over all relevant particle orientations?* Two orientation strategies are produced:

[1] make_figures.py


Habit shape files
adda_shape_files/<habit>/
(.adda + .meta.mat)
ADDA
(Discrete
Dipole
Approximation)
binary 1.2 /
1.4.0 / 1.5.0-
alpha3
Mueller matrix on
181x361 deg
scattering grid, per particle
orientation
adda_config_<habit>.py
frequencies, T,
tolerances,
symmetry flags
Python pipeline
adda_python_scripts/
orientation average +
spherical-harmonic
expansion
SSDB NetCDF
arts-sspdb-
compatible
(version
beta_0.9.1)
Radiative
transfer
ARTS / RTTOV

Strategy	Physical situation	Orientation space covered	Output coordinates
ARO — azimuthally random oriented	cloud particle whose symmetry/maximum axis stays **horizontal** (gravity), azimuth fully random	Euler-β tilt grid × incidence-angle grid (5 × 10 solved, re-gridded to 19 × 19 in output)	one NetCDF per (β, size, f, T)
TRO — totally random oriented	freely tumbling particle	full Euler space (α, β, γ) — integrated away	orientation-independent cross sections + Mueller

Part 0 — Orientation

0.1 Repository map

```
DDA/
+-- adda_python_scripts/        ← THE PIPELINE (orchestration, post-processing, export) -
 ↪ tracked
|                               in this repo with its full history since 2026-11-10 (report
 ↪ §13)
|   +-- adda_config_*.py         one config per habit (ALL knobs live here)
|   +-- adda_main_control.py     ARO entry point (T×f×size loop, job submission)
|   +-- adda_sub_control_spharmonics.py   one ARO job: icosphere orientation loop + ADDA
|   +-- adda_main_control_rndorient.py    TRO entry point (chunked job-table)
|   +-- adda_sub_control_randomorient.py  one TRO job: ADDA '-orient avg'
|   +-- adda_process_aro_particles*.py    post-processing (averaging, re-expansion)
|   +-- adda2SSDBnetcdf_functions.py      SSDB export writer
|   +-- definitions_for_netcdf.py         SSDB variable/dimension definitions
|   +-- adda_control_functions.py         (acf) ADDA CLI wrapper, dielectric, mirror
 ↪ transforms
|   +-- adda_orientational_avg_spharmonics_functions.py  (aoa) ARO orientation averaging
|   +-- spherical_harmonics_for_adda.py   (spha) shtns SH expansion machinery
|   +-- icosphere_grid.py        (ico) orientation-sampling grids
|   +-- old_configs/, *_for_testing.py, adda_sub_control_gridref.py  ← ARCHIVE, do not
 ↪ extend
+-- adda/                      ADDA 1.5.0-alpha3 (clone of m-brath/adda) ← NOT in git, §3.5
+-- adda_1.4/                  ADDA 1.4.0 ← PRODUCTION BINARY (Phase-3) ← NOT in git, §3.5
+-- adda_1.2/                  ADDA 1.2 (golden baselines made with it) ← NOT in git, §3.5
+-- adda_shape_files/            particle geometries:
 ↪ <habit>/Dmax…_Dveq…_Mass…_Freq…GHz.adda(+.meta.mat)
|                                ← NOT in git: external 1.8 GB input, provision a clone by
 ↪ hand (§1.2)
+-- adda_aux_input/              avg_params.dat files (TRO '-orient avg' integration
 ↪ control)
+-- arts-sspdb-interface/        upstream read-only tooling for the *published* ARTS
 ↪ database
|                                ← NOT in git: independent repo (m-brath fork), report §14
+-- SSD_scripts_deprecated/      deprecated SSDB-interface prototypes ← ARCHIVE (plain
 ↪ files since 2026-11-10)
+-- plans/                       modernization plan + running findings report (READ before
 ↪ touching code)
+-- baselines/                   frozen golden ADDA outputs + regression check tooling
+-- analysis/                    phase analysis/gate scripts + benchmark results
 ↪ (JSON/TXT/LOG)
+-- docs/                        this manual + figures + build scripts
```

Results never live in this repo. Intermediate and final data go to `<prefix_data>/adda_results/…` (cluster) or `<repo>/adda_results/…` (local mode). The old HPC results are *not accessible* — regression validation therefore relies on `baselines/` (golden ADDA outputs) and the equivalence suites in `analysis/`.

The geometry inputs are outside git as well. `adda_shape_files/` (1.8 GB) is untracked; a

fresh clone must be provisioned by hand before anything habit-related can run (§1.2).

0.2 Glossary

Term	Meaning in this project
ADDA	*Discrete Dipole Approximation* code (Yurkin & Hoekstra). Three trees in-repo: `adda_1.2`, `adda_1.4` (production), `adda/` (1.5.0-alpha3).
ARO / TRO	azimuthally random orientation (axis horizontal) / totally random orientation.
D_veq	volume-equivalent particle diameter; in result-folder names the field is spelled `deq` — and in **TRO** folders it misleadingly carries the equivalent **radius** (legacy, §D-14).
D_max	maximum particle extent (defines the size grid step naming, e.g. `Dmax00013um`).
size parameter x	$x = \pi D_{\mathrm{veq}}/\lambda = \pi D_{\mathrm{veq}} f/c_0$; `size_parameter_limit=10` caps the backlog.
dpl / dpl_{eff}	dipoles per (vacuum) wavelength — DDA resolution. `dpl_eff` is what the fixed shape file actually yields at the requested λ. ADDA's auto heuristic lands at $dpl_{\mathrm{eff}} \approx 18$ at production x.
resolution class / fmin	each physical size exists as several shape files tagged `frequency_min` $\in$ `{192, 337, 457, 671, 887} GHz`; the class decides `N_dipoles` (§1.2).
β **grid**	tilt of the particle symmetry axis from vertical — the ARO *output* coordinate (19 values in the SSDB grid). Not a solve axis.
incidence angle θ_{inc}	direction of the incoming plane wave relative to the *laboratory* z-axis — second ARO output coordinate (19 values). Not a solve axis either: ADDA solves per *icosphere vertex* (§1.5.2).
`-eps` **(ADDA)**	iterative-solver stop criterion: residual reduced by $10^{-\mathtt{eps}}$. `eps=3` production; noise floor §1.1.
`eps_sphexp`**=0.005**	per-vertex spherical-harmonic truncation error of the Mueller grid, relative to RMS of Mueller element M_{11} (§1.6).
`delta`**=0.01**	convergence tolerance of the icosphere orientation refinement loop (relative change of the averaged extinction matrices, §1.5.4).
`limit_sph`**=4**	max icosphere refinement level (levels: 12/42/162/642/2562 vertices; minimum 3 passes).
shtns	fast spherical-harmonics transform library (orthonormal, no Condon–Shortley phase here).
Matzler 2006	dielectric function of ice used for all ice habits: `acf.eps_ice_matzler06(f, T)`.
SSDB	ARTS Microwave Single Scattering Properties Database; our export format is `beta_0.9` with Phase-6 additive backscatter variables → current writer version `beta_0.9.1` (zenodo 1175572 lineage — Eriksson et al. 2018; ARO extension: Brath et al. 2020).

Term	Meaning in this project
backscatter matrix B **/** `backMat_data`	the exact 4×4 Mueller matrix at the *exact backscatter direction* (grid point Θ=180°, Φ=0° of each ADDA solve) — extracted raw, orientation-averaged and exported **outside** the spherical-harmonic route (Phase 6, §1.4); raw units are k-normalized, exported units m²/sr.
habit IDs	particle shape definitions from the ARTS habit catalogue: ID9 = Hong/MoCOLAG plate, ID7 = Hong column, ID20 = hexagonal plate aggregate, ID28 = Icon snowflake, ID3 = Liu sector snowflake.
golden / baseline	frozen ADDA outputs in `baselines/` used for regression (§4.1).

0.3 Status at the time of writing

- **Done (computed + processed):** ARO for `HexPlaAggCrystal_Id20`, `HongPlate_Id9` (results on HPC, currently not accessible), plus analytic/voxel verification cases.
- **Backlog ready** (configs verified, jobs counted, not yet run): ARO `HongPlate_Id9` full 35-channel grid (5235 jobs), `HongColumn_Id7` (1524), `IconSnow_Id28` (1276), `LiuSectorSnowflake_Id3` (1252); TRO `HongPlate_Id9_TOR` (4608 jobs). Estimated ≈ 2700 single-core CPU-h total (see §4.4).
- **Phase 6 done (2026-11-06):** the exact 4×4 **backscatter Mueller matrix** (ADDA `mueller_scatgrid` Θ=180°/Φ=0° slice) is extracted per vertex, cached, mirror-filled, orientation-averaged and exported as `backMat_data`/`backMat_index` — SSDB writer version **beta_0.9.1** (§1.4, §1.8; gate suite `analysis/phase6_backscatter.py`).
- **Phase 7 done (2026-11-06):** the internal per-run Mueller artifact is a single **NetCDF4** file `<shape>_mueller_data.nc` (replaces `.pkl`, which replaced ARTS-XML); all 12 local legacy artifacts migrated bitwise (§2.3; report §7).
- **Phase 8 done (2026-11-07):** size-parameter-adaptive scattering grid (`scatgrid_adaptive=True` config switch — still OFF in production habit configs; laws + gate twins in report §8) and export-side SH cap (`export_sph_margin=10`, 15× smaller beta files) (§1.6.1).
- **Phase 9A done (2026-11-09):** analytic diffraction-peak model scanned (analysis-only): **conditional GO** for a gated model+residual hybrid — big spectral savings where the eikonal model is valid (face-on cells), actively harmful at grazing incidence. Nothing wired into the pipeline yet (§1.9; report §9).
- **Production ADDA binary: 1.4.0** (locally built; equivalence-gated against 1.2 and 1.5.0-alpha3, §4.2). Cluster branch configs still default to 1.2 until an MPI 1.4 build exists there — override with `ADDA_BINARY`/`ADDA_VERSION` (§3.5).
- Before touching **any** pipeline code, read `plans/DDA-modernization-report.md` (§1.5 breakages, §1.6 latent-bug catalogue, Phase 2–7 sections list what is deferred).

Part 1 — Theory

1.1 The Discrete Dipole Approximation (DDA)

The DDA represents a particle of arbitrary shape as a finite array of **N polarizable dipoles** sitting on a cubic lattice with spacing l. Each dipole acquires a dipole moment $\mathbf{P}_i = \alpha_i \mathbf{E}_{\text{tot}}(\mathbf{r}_i)$, and the total field is the incident field plus the field scattered by *all* dipoles — which yields a dense linear system

$$\mathbf{E}_{\text{tot}}(\mathbf{r}_i) \;=\; \mathbf{E}_{\text{inc}}(\mathbf{r}_i) \;+\; \sum_{j \neq i} \mathbf{A}_{ij}\, \mathbf{P}_j, \qquad i = 1, \ldots, N$$

solved iteratively (ADDA default: GMRES stabilized with the Lanczos-type solver family) rather than by direct factorization. The particle shape is encoded simply by *which* lattice sites carry dipoles — this is why arbitrary habits (aggregates, snowflakes, columns) are tractable, and why the whole pipeline can be driven purely from `*.adda` position files.

What the maintainer must know about ADDA's numerics (all measured in `analysis/phase4b_adda_benchmark.py`, data in `analysis/phase4b_bench_results.json`, figure below):

Fact	Consequence for this project
`-eps <n>` = iterative solver stops at residual 10^{-n}. It controls **solver** error only.	`eps=3` production: own-solver scatter is $\leq 10^{-11}$ for small jobs but $\sim 8 \cdot 10^{-4}$ at $x = 5$ — so **identical ADDA re-runs are not bit-reproducible**; every regression gate must be noise-based, never bit-level.
Error budget is **discretization-dominated**: at $x = 5$ going dpl_{eff} from 4.5 to 7.5 moves Cext by $2.5 \cdot 10^{-1}$ → ref-value, while $\varepsilon_{\text{solver}}$ 4→2 moves it only $\sim 1.5 \cdot 10^{-3}$.	Tuning `-eps` below 3 buys nothing; resolution (the shape file class) is the lever (§1.2).
ADDA's auto dpl heuristic lands at $dpl_{\text{eff}} \approx 18$ at production $x \in (1, 8)$.	The fixed per-file dipole counts are deliberately over-resolved; this is a *design choice* (one geometry file reused across all frequencies of its class).
Cost per solve: $N = 12\,312$ **dipoles → 0.4 s,** $N = 48\,597$ → **9.5 s** (single core, ADDA 1.4.0, macOS). Scaling $\propto N^{2.3}$.	Backlog cost model §4.4; also why solves are embarrassingly process-level parallel (run N single-core ADDA calls per job, §3.6).
Cext/Cabs from a DDA run are only as good as the shape discretization; mirrors/rotations of the *voxel set* are exact lattice operations (§1.7).	Symmetry flags must be verified on the actual voxel set, not on the nominal habit.

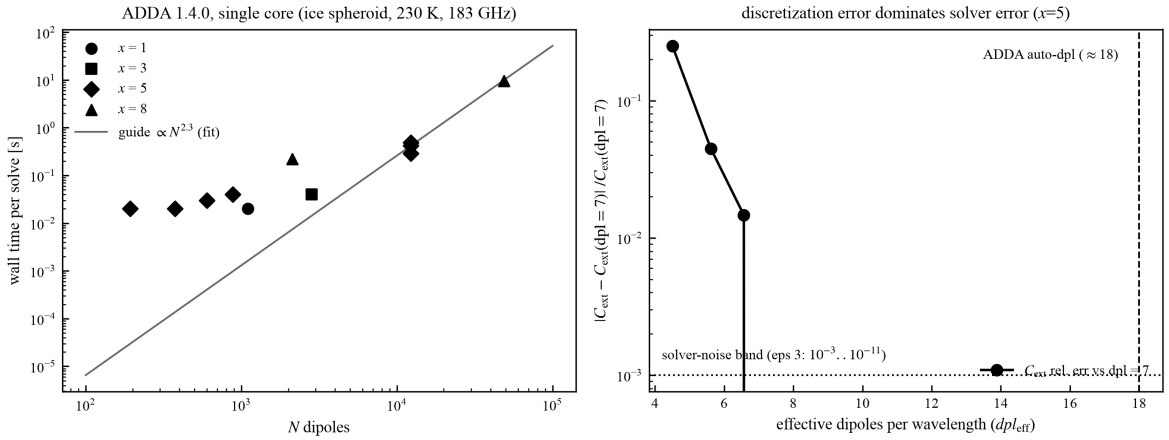


Fig. 4 — one ADDA solve: cost and error budget. *Left panel: wall time grows like $N^{2.3}$ with dipole count — the cost-planning input. Right panel: at $x = 5$, the Cext error from dipole resolution (dpl_{eff}, x-axis) stays 2–3 orders of magnitude ABOVE the solver-noise band (red dotted line, `-eps 3`): the shape-file resolution class — not `-eps` — is what sets accuracy. (Data: oblate spheroid 1.0/0.5, Matzler ice 230 K, 183 GHz, `analysis/phase4b_bench_results.json`.)*

Key references: Yurkin & Hoekstra 2011, JQSRT 112, 2234–2247; ADDA manual `adda/doc/manual.pdf` and `adda_1.2/doc/manual.pdf`.

1.2 Particle shape files and the resolution scheme

Shape model and file contract

Every geometry is a pair of sibling files in `adda_shape_files/<habit>/`:

```
Dmax00013um_Dveq00010um_Mass4.79983e-13kg_Freq0192.000GHz.adda      ← dipole positions
 ↪  (x y z, one per line, lattice units; 1 header line)
Dmax00013um_Dveq00010um_Mass4.79983e-13kg_Freq0192.000GHz.meta.mat  ← MATLAB metadata
 ↪  struct 'meta'
```

The `.meta.mat` (read with `acf.loadmat`) drives *all* size/frequency bookkeeping. Fields actually consumed by the pipeline: `diameter_vol_eq` [m], `diameter_max` [m], `frequency_min` [Hz], `N_dipoles`, `mass` [kg], `lambda_dpl` (=λ at fmin, m — used to fill the SSDB `dpl` field), plus provenance (`habit_id`, `source`, `medium`, …). **A `.adda` without its `.meta.mat` sibling is invisible to the pipeline** (the configs glob `*.mat` for the size tables).

The model is **cubes on a cubic lattice**. Mirrors map the lattice onto itself — which is exactly what the symmetry-flag tests of §1.7 exploit. Rotations by non-multiples of 90° never do, so a voxelized hexagon is *never* an exactly C6-symmetric model; ideal-shape C6 exists sub-voxel only, and exploiting it would cost a boundary-layer discretization error. This is why rotation symmetries (`sym_Zobj`) are parked while mirror symmetries are exploited.

Shape-data provisioning (this folder is NOT in git)

`adda_shape_files/` is untracked and gitignored: 1.8 GB of input geometry (970 `.adda` + 970 `.meta.mat`, six habit folders). It was committed in the initial snapshot and **purged from the entire repository history on 2026-11-10** — it was 95 % of the object store and its largest blob (170 MB) exceeded GitHub's 100 MB push limit (findings + gate table: report §10).

- **A fresh clone has no geometry.** Every habit config, `analysis/inventory.py`, `baselines/make_baseline.py` and the gate suites glob this folder, so nothing that touches a habit runs until it is populated — `git clone` alone is not enough.
- **Copy the tree** into `<repo>/adda_shape_files/` keeping the layout `<habit>/Dmax…_Dveq…_Mass…_Freq…GHz.adda` + sibling `.meta.mat` (any missing sibling makes that size invisible to the pipeline, see above). Authoritative sources, in order of trust:

1. the pre-purge git bundle `../DDA-prepurge-2026-11-10.bundle` — byte-verified archive of all 1940 files: `git fetch <bundle> refs/heads/main:refs/heads/prepurge-backup` then `git archive prepurge-backup adda_shape_files | tar -x -C .` (then drop the temp ref);
2. the HPC working copy under the cluster prefix (Mistral/Thunder — confirm the exact path before relying on it);
3. this checkout's own working tree.

- **Verify after provisioning:**

```
find adda_shape_files -type f | wc -l   # expect 1940 = 970 .adda + 970 .meta.mat
```

```
python analysis/inventory.py          # habit×size×frequency table must be non-empty
```

- **Do not put it back into git** (`git add -f` defeats the ignore rule and reintroduces the unpushable 170 MB blob). If the geometry ever needs versioning, use a separate data repo or a citable archive (zenodo), not this history.

The 5-resolution scheme

Each physical size ships as several shape files tagged with `frequency_min`:

- classes: **{192, 337, 457, 671, 887} GHz**;
- `N_dipoles` is **fixed per file** — ADDA is *never* asked to resample; it rescales the geometry (`-eq_rad`) so the same dipole array represents the particle at any λ;
- the effective resolution at frequency f is therefore $dpl_{\text{eff}} = dpl_{\text{design}} \cdot (f_{\min}/f)$ — i.e. a file only gets *finer* than designed when used below its fmin.

Two regimes exist in practice (measured on `HongPlate_Id9`, fig. right panel):

particle	classes	N_dipoles across classes
small (10 μm)	5	**identical** (1015 in all five — degenerate copies: even the coarsest lattice already exceeds the needed dpl)
mid (341 μm)	5	1258 for classes 192/337/457, then 2740, 6255
large (2596 μm)	5	27 644 → 165 072 → 379 780 → 1 184 589 → 2 956 210 ($\approx f_{\min}^3$)

Also note: **class availability can be per-size incomplete** (e.g. HexPlaAgg 197 μm has only the 671 and 887 classes), so never assume five files per size.

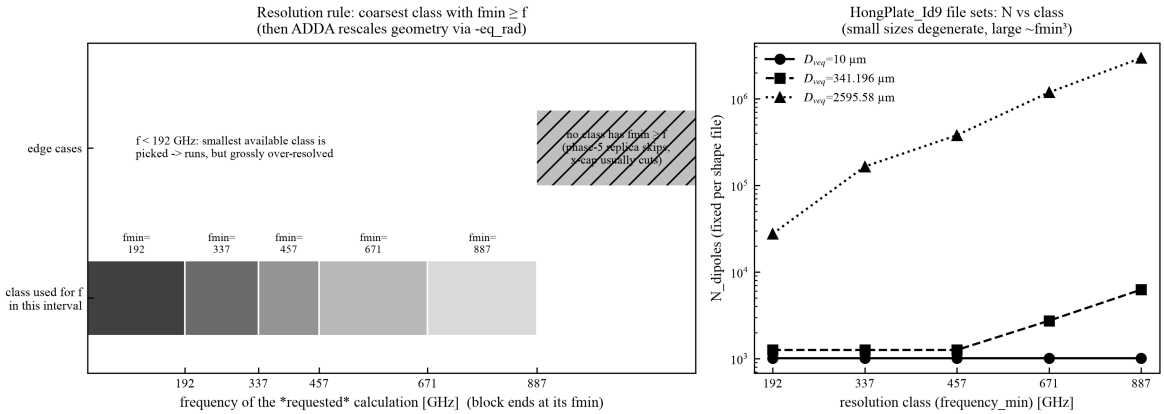


Fig. 3 — Which shape file computes a given (size, f). *Left: a requested frequency falls in one grey block (one per `fmin` class); that block's `fmin` class file does the run (block label sits at its right edge). The two edge cases are annotated. Right: measured N_dipoles per class for three real HongPlate sizes — small particles are degenerate (one lattice already suffices), large ones grow* $prop to f_{min}^3$.

The selection rule (as implemented in `adda_main_control.py`)

For each (size, f): among the files of that physical size, the code picks the **coarsest class with fmin ≥ f** (`df = fmin−f; df[df<0]=inf; argmin`); every *other* file of that size prints "wrong particle realisation" and

skips. Consequences, precisely:

- f inside a class interval ($f_{\min,k-1}$, $f_{\min,k}$] → class k computes it ($dpl_{\text{eff}} \geq$ design value — always on the safe/over-resolved side);
- **f below all classes** (e.g. the 1–175 GHz entries of the frequency list) → the *smallest available class* is picked and the job **does run**, grossly over-resolved;
- **f above all classes** → with `df` all-∞ the raw control code would accidentally pick the first file in folder order; the analysis replicas (`phase5_backlog_inventory.py`) additionally guard `if not np.isfinite(df).any(): continue`. In production the x-cap removes these combinations anyway (HongPlate's 886 GHz < 887 class).
- The size cap: $x = \pi D_{\text{veq}} f / c_0$ > `size_parameter_limit` (= 10) → skip (146 of HongPlate's 8925 (size,f) combos; the *majority* of combos are killed by resolution bookkeeping, not by x — 7701 by the fmin rule; report §1.7).

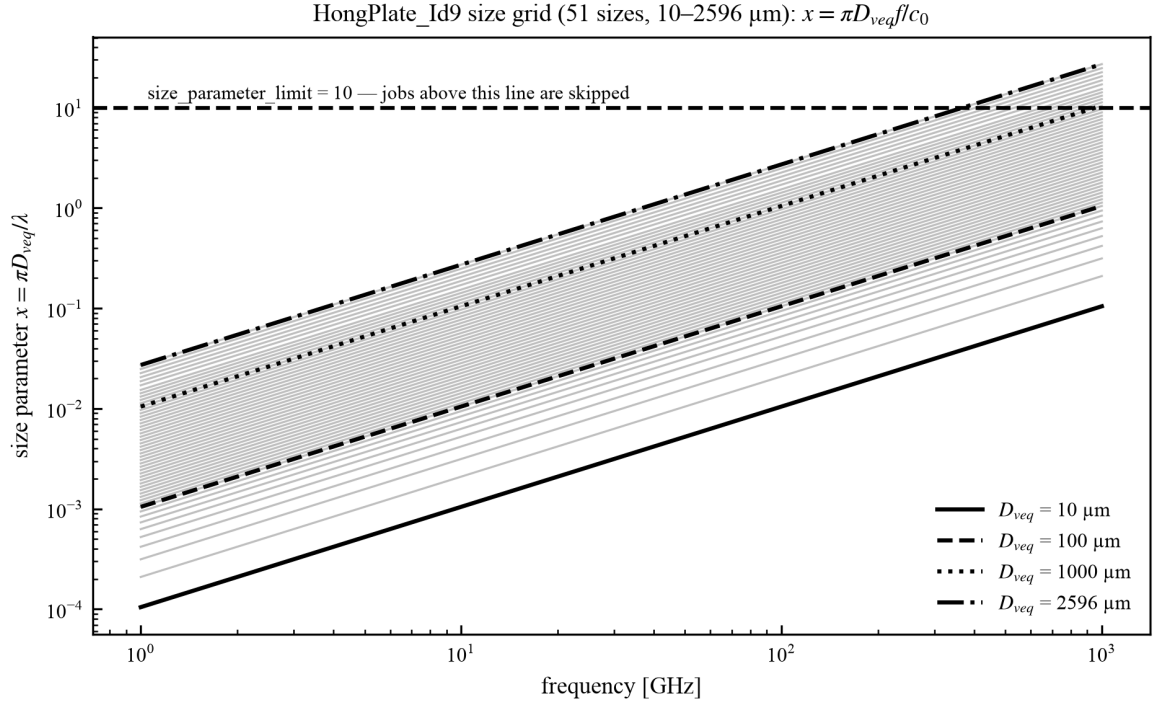


Fig. 2 — What the x-cap actually removes. *Each line is one particle size ($x = piD_{veq}f/c_0$ proptoDf, straight in log-log). Everything above the dashed $x = 10$ line is skipped: for HongPlate that is 146 of 8925 (size, f) combinations — far fewer than the 7701 killed by the resolution rule of Fig. 3.*

1.3 Dielectric function of ice (Matzler 2006)

All ice habits use the Mätzler (2006) parametrization, implemented in `acf.eps_ice_matzler06(f, TK)` (`adda_control_functions.py:1068-1103`):

$$\varepsilon(f,T) \;=\; 3.1884 + 9.1\cdot 10^{-4}\,(T-273) \;+\; i\left[\frac{\alpha(T)}{f} + \beta(T)\,f\right], \qquad f \text{ in GHz},\ T \text{ in K}$$

Valid for 0.01–3000 GHz, 20–273.15 K (raises otherwise). **It returns the complex permittivity, not the refractive index** — callers compute $m = \sqrt{\varepsilon}$ and hand `-m m_re m_im` to ADDA. Note the absorption is strongly frequency-dependent (loss $\propto 1/f$ at low f, $\propto f$ at high f) and strongly temperature-dependent: the imaginary part at 190 K vs 270 K differs by orders of magnitude — hence the three production temperatures

{190, 230, 270} K are *not* a formality; each (T, f, size) is a separate ADDA job.

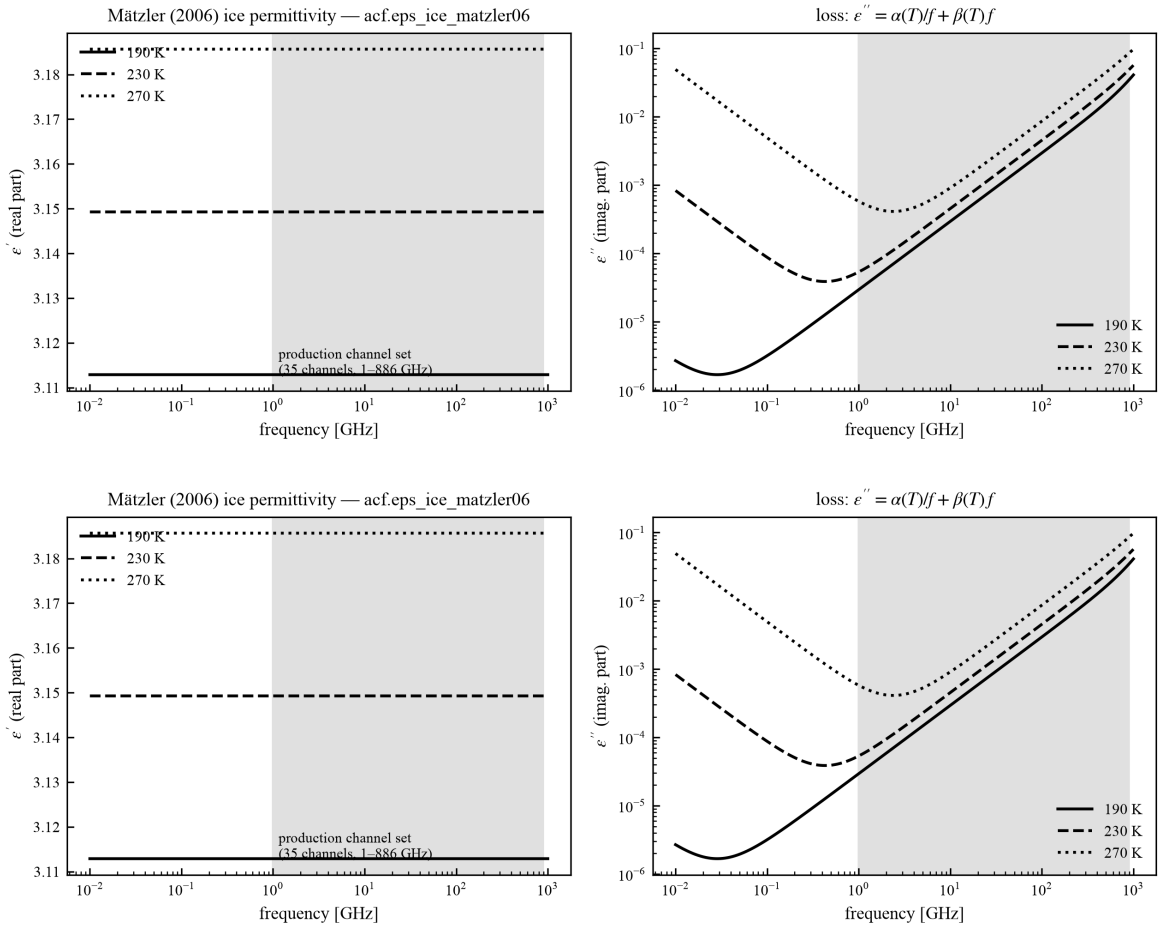


Fig. 1 — Mätzler ice permittivity across the whole valid range. *Grey band: the production channel set (35 frequencies, 1–886 GHz). Reading: the real part (left) barely moves (≈3.11–3.19, only with T); the imaginary part (right) spans ~4 orders of magnitude in f and T — the loss tangent at mm/sub-mm wavelengths is what makes cloud ice a microwave-radiometry issue at all, and why each (T, f) pair is its own ADDA job.*

1.4 From ADDA output to Mueller and extinction matrices

Each ADDA solve produces two parsed artifacts (parsing in `acf.adda_call`, `adda_control_functions.py:198-236` — **positional, column indices hardcoded**; re-verify when switching ADDA versions, report §1.3):

1. `ampl_scatgrid` — the complex *amplitude matrix* S(f^) on the scattering grid. Row 0 = forward direction (θ=0). Column layout (real/imag pairs) is **not** in textbook order:

cols	quantity
0, 1	s_{22}
2, 3	s_{11}
4, 5	s_{12}
6, 7	s_{21}

From the forward amplitudes the **extinction matrix** (4×4, units μm^2) is built via the Mishchenko (2002)

optical theorem construction — amplitudes scaled by $\lambda/(-2\pi i)$, then $K\cdot = \lambda$ (`acf.extinction_matrix`, :40–80). *The ARO pipeline's C_ext comes entirely from this grid column — which is why `-Csca` and `-alldir_inp` were removed from all ARO configs (Phase 3): nothing ever parsed them.*

2. `mueller_scatgrid` — the 16 Mueller-matrix elements M_{11}…M44 (row-major) on the scattering grid from `scat_params.dat`: **181 θ (0–180°) × 361 φ (0–360°), 1° resolution**; col 0 = θ, col 1 = φ (degrees). This is the *angular scattering signature of one fixed particle orientation* — the object that §1.6 expands in spherical harmonics. The θ/φ here live on a grid **attached to the propagation direction** — a crucial fact for the mirror transforms of §1.7 (θ is never remapped).
3. **Backscatter slice (Phase 6)** — from the *same* `mueller_scatgrid`, `acf.adda_call` additionally returns

- `back_matrix`: the 4×4 element vector at **Θ=180°, Φ=0°** (exact backscatter), and
- `back_mat_ring`: the full Φ ring at Θ=180° (needed for the mirror fill-in and the convention analysis, §1.7).

These are **raw, k-normalized** values (dimensionless $|f|^2$-like — *not* µm²/sr; the unit conversion $\lambda^2/4\pi^2$ happens only in `rawdata2average`). `adda_call` raises if the scat grid has no Θ=180 row or Φ=0 column — the production 181×361 grid has both. This exact-direction matrix travels through the stack **outside** the spherical-harmonic route: Brath et al. (2020) flag SH phase functions as unreliable at the exact backscatter direction, so the exported `backMat_data` (SSDB, §1.8) is the trustworthy value there. The orientation averaging uses `aoa.pol_changer_back` with the convention $B_{\text{lab}} = R(+\varphi_{\hat{n}})\, B_{\text{raw}}\, R(+\varphi_{\hat{k}})$ — the *opposite* rotation sign to the extinction branch (the scattered Stokes basis is attached at $-\hat{v}$; pinned by gate scan, report §6.2: every alternative convention scores ≥31× worse).

The two output flags `-scat_matr both` request amplitude + Mueller; `-store_scat_grid` + `-scat_grid_inp scat_params.dat` request the grid itself. The geometry of *both* angular coordinate systems — the lab-frame incidence/tilt angles and the propagation-attached scattering grid — is shown below:

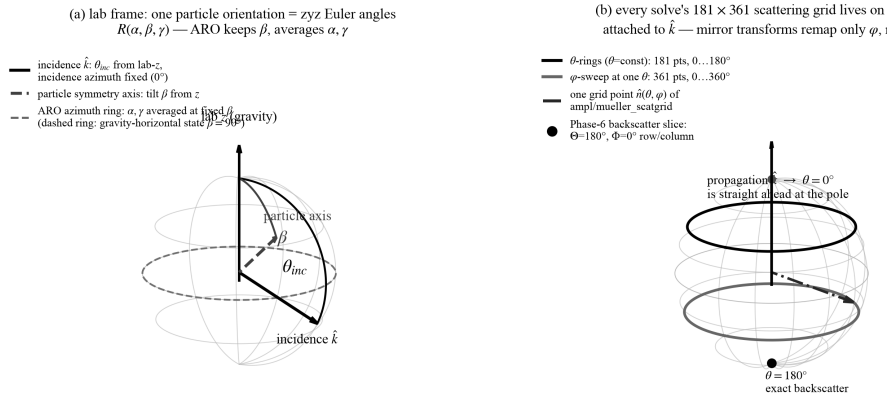


Fig. 8 — the two coordinate systems you must not confuse. *(a) lab frame: the incidence direction $\hat{k}$ (θ_{inc} from lab-z, azimuth fixed) and the particle symmetry axis (tilt β); ARO averages the azimuth/spin Euler angles α, γ at fixed β. (b) every solve's* 181×361 *amplitude/Mueller grid lives on a sphere **attached to** $\hat{k}$ — $\theta = 0°$ is the forward pole, and the filled black pole ($\Theta = 180°$, $\Phi = 0°$) is the Phase-6 backscatter slice. This is also why the §1.7 mirror transform remaps only φ.*

1.5 Orientation averaging — the two strategies

1.5.1 Why averaging, and in which coordinates

A real cloud contains particles in many orientations. The observed scattering is the orientation average of the single-particle Mueller matrix. Euler angles are taken in ADDA's **zyz convention** (α about lab-z, β about the line of nodes, γ about the particle's own z), i.e.

$$R(\alpha,\beta,\gamma) \;=\; R_z(\alpha)\, R_{z'}(\beta)\, R_{z''}(\gamma), \qquad \text{ADDA flag } \texttt{-orient}\ \alpha\ \beta\ \gamma\ \text{[deg]}$$

> **Source methodology (ARO):** the TRO/ARO classification, the zyz Euler convention, the α/γ-averaging at fixed tilt β, and the resulting ARO dependence on (β, θ_{inc}, scattering angles) follow **Brath et al. (2020, AMT 13, 2309–2333)** — the foundation paper of this project's ARO approach (see References). Their hexagonal-plate / plate-aggregate ARO database itself builds on the shape and frequency/temperature setup of **Eriksson et al. (2018, ESSD 10, 1301–1326)**, which is also where the habit IDs used in this repo (`HongPlate_Id9`, `HongColumn_Id7`, …) originate.

	ARO	TRO
free orientation parameters	tilt β of the particle axis; azimuth (α, γ) averaged	all of α, β, γ
physically motivated state	axis horizontal: $\beta = 90°$ (gravity-brained crystals)	freely tumbling particle
what is produced	the full grid $\beta \in [0°, 180°]$ (19 values) — users weight the tilt distribution downstream	orientation-independent result (integrated inside ADDA)
weighting	uniform in α, γ at each β	uniform over the rotation group

The SSDB represents the ARO state on a 2-D output grid (β, θ_{inc}): β = tilt of the particle axis from vertical (0–180°, 19 output values), θ_{inc} = angle between the incoming plane wave and the lab-z axis (0–180°, 19 output values, azimuth of incidence fixed to 0 — the remaining azimuthal randomness is what makes it "azimuthally random"). The α and γ integrations are performed numerically (Gauss–Legendre) at every (β, θ_{inc}) node.

> **Critical design fact (Phase-4b, report §4b.1):** the β and θ_{inc} grids are **SSDB output coordinates, not quadrature axes**. The number of ADDA *solves* is set solely by the icosphere orientation-vertex grid (§1.5.2) — changing `number_of_betas` or the incidence-angle resolution does not save or cost ADDA solves. And the tempting "half-range reuse" symmetry (solve β<90° and mirror to β>90°) is **provably wrong** (measured discrepancies $2.3\cdot10^{-3}\ldots7.2\cdot10^{-2}$ vs 10^{-11} solver noise, `analysis/phase4b_halfrange_check.py`).

1.5.2 ARO sampling: the icosphere vertex grid

One ARO job (one T, f, size) samples particle orientations as the **vertices of a refined icosahedron** on the unit sphere. A vertex *is* the incidence direction: ADDA is called with `-prop vx vy vz` (fixed particle attitude `-orient 0 0 0`, beam coming from the vertex direction). Vertex counts per refinement level L:

$$V(L) = 10\cdot 4^L + 2 \;:\quad 12,\ 42,\ 162,\ 642,\ 2562, \ldots \qquad (\text{triangles } 20\cdot 4^L)$$

per vertex: ADDA solve → parse Mueller grid → SH-expansion (§1.6) → per-vertex NetCDF cache. The full $\alpha/\beta/\gamma$ machinery then reconstructs the orientation average by rotating this vertex-resolved data set

(`ico.rotate_vertices` + barycentric interpolation on the rotated icosphere + Stokes polarization-basis rotation `aoa.pol_changer`).

1.5.3 ARO averaging quadrature

Inside `adda_sub_control_spharmonics.py` (:629–814) the **extinction matrix** is averaged at each $(\beta, \theta_{\text{inc}})$ node with Gauss–Legendre (GL) quadrature in α and γ:

$$n_{\text{GL}} = 4 \cdot 2^{L} \qquad \text{GL nodes mapped to } [-180^\circ, +180^\circ] \text{ in } \alpha \text{ and } \gamma,$$

for every β (from `number_of_betas`, 5) and θ_{inc} (`incidence_angles`, 10 values): rotate the vertex set by $(\alpha, \beta, \gamma) \rightarrow$ barycentric interpolation of the extinction matrix at the incidence direction $\rightarrow$ Stokes polarization-basis rotation $\rightarrow$ GL-weighted sum over γ, then over α.

The **Mueller** SH-coefficient averaging happens later, in post-processing (`aoa.rawdata2average`, joblib-parallel, output re-gridded to the 19 × 19 SSDB grid).

1.5.4 Convergence of the orientation sampling

The icosphere level L is refined until the averaged extinction matrices stop changing:

$$\varepsilon_{\text{rel}} = \frac{\|\overline{K}^{(L)} - \overline{K}^{(L-1)}\|_2}{\|\overline{K}^{(L-1)}\|_2} \leq \delta \ (= 0.01), \qquad \overline{K} \in \mathbb{R}^{n_\beta \times n_\theta \times 4 \times 4}$$

at least 3 passes ($L = 0, 1, 2$), at most `limit_sph` ($= 4$).

Note the two *different* accuracy controls and what each drives:

knob	drives	tolerance
`delta`	icosphere **orientation** refinement (# ADDA solves)	0.01 on averaged extinction
`eps_sphexp`	per-vertex **spherical-harmonic truncation** of the Mueller grid	0.005 rel. to M_{11} RMS
`-eps`	ADDA internal solver residual	10^{-3} (noise floor, §1.1)

1.5.5 TRO: two strategies, one decision

strategy	mechanism	yields	status
ADDA built-in `-orient avg <avg_params.dat>`	Romberg integration over α,β,γ *inside ADDA*; writes integrated `mueller` (2° θ-grid) + `CrossSec*` files	cross sections **and** orientation-averaged Mueller	**production for TRO** (Phase-5 decision, report §5.1)
own sampling + SH expansion (icosphere, as ARO)	same machinery as ARO but full Euler coverage	Mueller grids → SSDB	reserved for hypothetical TRO *grid* products; not needed for the backlog

Hard constraint discovered in Phase 3: **ADDA 1.4/1.5 reject `-orient avg` combined with `-scat_grid`, `-Csca` or `-alldir`** — built-in averaging never produces *per-angle* scattering grids (the integrated mueller

is a separate product). The avg file (zyz, per-angle Romberg control `Jmin/Jmax/eps/equiv/periodic`) for the production habit — `adda_aux_input/HongPlate_Id9_TOR_sizegrid/avg_params.dat`:

Euler angle	range	$J_{\min}/J_{\max}$	tol.	equiv	periodic	note
α	$0\ldots360^\circ$	2 / 5	0	true	true	periodic → trapezoid + Richardson
β	$0\ldots90^\circ$	2 / 4	10^{-6}	false	false	**XY-mirror fold** (§1.7!)
γ	$0\ldots360^\circ$	2 / 4	10^{-6}	true	true	

Folding β to 0–90° is *only legitimate because the voxel set is XY-mirror closed (closure 1.000 on the TOR grid).* It was **gated** against the full-range file (`avg_params_fullrange.dat`): CrossSec rel-diff $\leq 1.35\cdot10^{-5}$ (= the Romberg tolerance itself), Mueller $\leq 10^{-13}$ → the fold is exact to integration accuracy (`analysis/phase5_tro_beta_gate.py`, PASS, report §5.3).

1.6 Spherical-harmonic expansion over the scattering angles

Every ADDA solve returns the Mueller grid $M_1,\ldots,M_{16}(\theta,\varphi)$ on the scattering grid — 181×361 static in production, or the size-parameter-adaptive grid of §1.6.1. Rather than storing 130k floats × 16 per vertex, the pipeline stores a **spectral representation** per vertex (`spherical_harmonics_for_adda.py`):

- transform: `shtns.sht(ntrunc, ntrunc, 1, sht_orthonormal | SHT_NO_CS_PHASE)` — orthonormal, no Condon–Shortley phase; input field interpolated from the θ-grid onto Gaussian latitudes;
- **error measure** — the L^2 sphere norm of the analysis→synthesis round-trip residual of the truncated expansion:

$$\delta[f] \;=\; \sqrt{\iint |f_{\text{rec}}(\theta,\varphi) - f(\theta,\varphi)|^2 \,\sin\theta\;\mathrm{d}\theta\,\mathrm{d}\varphi}\,,$$

per Mueller element the absolute target is ε_m = RMS(M_{11})·`eps_sphexp`; the returned relative error is $\delta[M_m]/\,\mathrm{RMS}(M_{11})$; - if the round-trip passes at full truncation, a **binary search** finds the *smallest* `ntrunc` still meeting `eps_m` — coefficient counts are per-vertex and adaptive; - coefficients are stored as **complex64** (this sets the floor of export precision; `map_sphexp/reshape_sphexp_2_common_exp` later remap every (β,θ_{inc}) node's coefficient set into one common shtns layout).

Why expand over scattering angles at all? Because the orientation average at a rotated orientation is then a **Wigner rotation of coefficients**, and phase-matrix products needed by radiative transfer stay fast; the post-processing additionally *re-expands* the averaged phase matrix on the fly to `eps_phase=0.001` on the 19 × 19 grid (§3.3 step 2).

1.6.1 Adaptive scattering grid (Phase 8)

The static 181 × 361 grid grossly over-samples small-x solves. With `scatgrid_adaptive=True` (config switch, §3.1; **still OFF in production habit configs** — probe configs `adda_config_phase8_hp_{adaptive,static}.py` only) the per-run `scat_params_adaptive.dat` is generated from the size parameter $x = \pi D_{veq}/\lambda$:

- **phi:** even periodic ring, `nphi = 2·ntrunc_cap + 2`, endpoint **excluded** ($\varphi_{max} = 360 - 360/n\varphi$); `ntrunc_cap = ceil(6·(x+4x^{1/3}+2))` — the scattering grid must accommodate $2\,n_{\text{trunc}}$ longitudes (§1.9.3 for why $n_{\text{trunc}} \propto x$). A duplicate $\varphi = 0/360$ row injects FFT aliasing error $\propto \Delta\varphi^2$ (measured $3.2\cdot10^{-2}$ twin failure), and odd rings break the Phase-4 mirror φ-index map (180° must be a ring node);

- **theta:** $n_\theta \le \min(181, \max(\text{cap} + 2,\ \pi x/0.1 + 1))$; the exact $\theta = 0$ and $\theta = 180$ rows are always kept (forward extinction parse, Phase-6 backscatter slice);
- `spha.sph_expansion` initial truncation is $(n_{lons} - 1)//2$ (shtns needs $n_\varphi > 2m_{max}$ on even rings; identical value 180 for the static grid);
- provenance in overview mat + nc root attrs (`size_parameter`, `scatgrid_ntrunc_cap`, `scatgrid_adaptive`); measured per-solve speedups ~8–17× at small x, ~2× at $x = 5$;
- **export side:** the SSDB writer additionally caps the *stored* SH dimension at `acf.scatgrid_ntrunc_cap(x, margin=export_sph_margin=10)` — beta files became **15× smaller** with bitwise-identical kept content (gate `analysis/phase8_export_twin.py`, report §8.8).

1.7 Mirror-symmetry exploitation (Phase 4)

If the **voxel set** is invariant under reflection at a coordinate plane of the particle frame, the Mueller grid of the mirrored orientation is obtained from the solved one by an exact index/sign transform — the ADDA solve is skipped and the vertex is *filled*.

Verified transform (`acf.MIRROR_*`, `adda_control_functions.py:1252-1289`; derived empirically, `analysis/phase4_grid_symmetry.py`, residual $\le 3 \cdot 10^{-11}$ at `-eps 3`):

mirror (plane)	axis flipped	scattering-grid remap	Mueller element signs
XY ($z \to -z$)	z	$\varphi \mapsto 180^\circ - \varphi$	for an odd number of mirrors: element-wise sign by $\mathbf{d}\mathbf{d}^\top$ with $\mathbf{d} = (1, 1, -1, -1)$ (outer product on the Stokes row/col pairs)
XZ ($y \to -y$)	y	$\varphi \mapsto -\varphi$	
YZ ($x \to -x$)	x	$\varphi \mapsto -\varphi$	(even number of composed mirrors → identity sign)

θ is never remapped — `mueller_scatgrid` lives on a propagation-*attached* grid; only φ moves. Fill-in mechanics in the sub-control (:265–322, :414–419): vertices on the negative side of an enabled plane are solved only if no fill exists yet (`_runfirst` ordering guarantees every mirrorable vertex is filled before it is reached; a missing fill can never silently produce NaN); the image vertex is found by nearest-vertex lookup (warns if snapping exceeds 10^{-3} rad); the filled grid is re-expanded in SH and cached like a solved vertex.

Backscatter ring (Phase 6): the Θ=180° Φ-ring of a skipped vertex is reconstructed from the source ring by the *same* φ remap times the elementwise sign rule $\mathbf{D}\mathbf{D}^\top$, $\mathbf{D} = (1, 1, -1, -1)$; the back matrix itself is the Φ=0° element of the filled ring. Verified **exactly** on directly-computed XY twin vertices (filled vs computed agree to 10^{-25} absolute — dropping the sign rule leaves $2 \cdot 10^{-13}$ at a $3.6 \cdot 10^{-11}$ scale). Note $\text{ring}(180^\circ, 180^\circ) = \text{ring}(180^\circ, 0^\circ)$ (basis π-rotation = conjugation by $-\mathbf{I}$), so the elementwise `M**Nsym` rule is exact even for XY (whose φ map sends 0°→180°).

Precondition — voxel closure. A flag is legal only if the *voxel set itself* is mirror-closed, and closure is **resolution-dependent** — measured by `analysis/phase4_symmetry.py` (figure): a fill across a non-closed mirror is a real **shape error** (HongPlate coarse-class YZ: 0.986 closure → $1.5 \cdot 10^{-4}$ Mueller field error vs $4.5 \cdot 10^{-12}$ when closed), not solver noise. Hence flags must be set from the **coarsest resolution class** of the size grid.

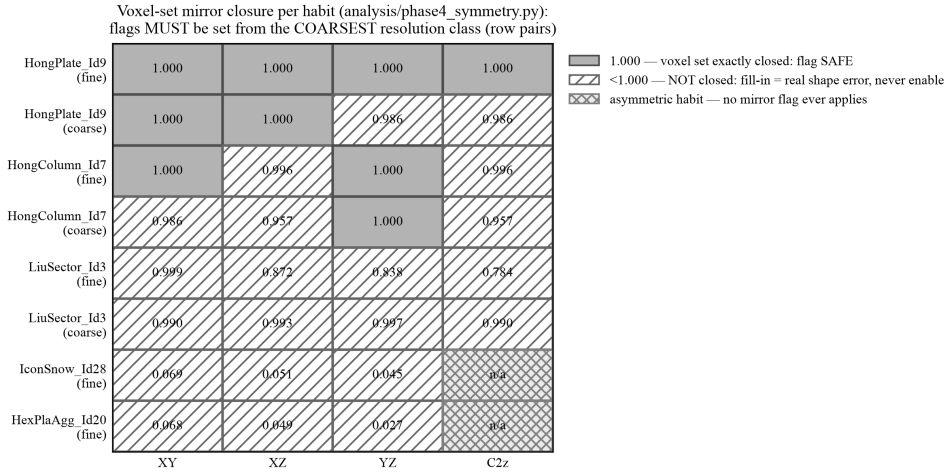


Fig. 6 — the mirror-flag decision table (fill texture = the rule itself). *Solid grey: voxel set exactly closed (1.000) at that resolution → flag may be enabled. Diagonal hatch: <1.000 → the fill-in would reconstruct voxels that do not exist = a real SHAPE error, never enable — note how a mirror can be closed at fine resolution but broken at the coarsest class, which is why flags follow the coarsest row of each pair. Cross-hatch: habit intrinsically asymmetric.*

Production flags (report §4.7): **HongPlate_Id9 + TOR grid: XY+XZ=true, YZ=false**; **HongColumn_Id7: YZ=true**; LiuSector/Icon/HexPlaAgg: none (aggregate/snowflake shapes break all mirrors). Measured effect (§4.8):

habit	solves	wall
ellipsoid gate (3 mirrors)	162 → **27** (+135 fills)	3.0×
HongPlate_Id9 (XY+XZ)	642 → **177** (+465 fills)	1.73×
HongColumn_Id7 (YZ)	642 → **337** (+305 fills)	1.39×

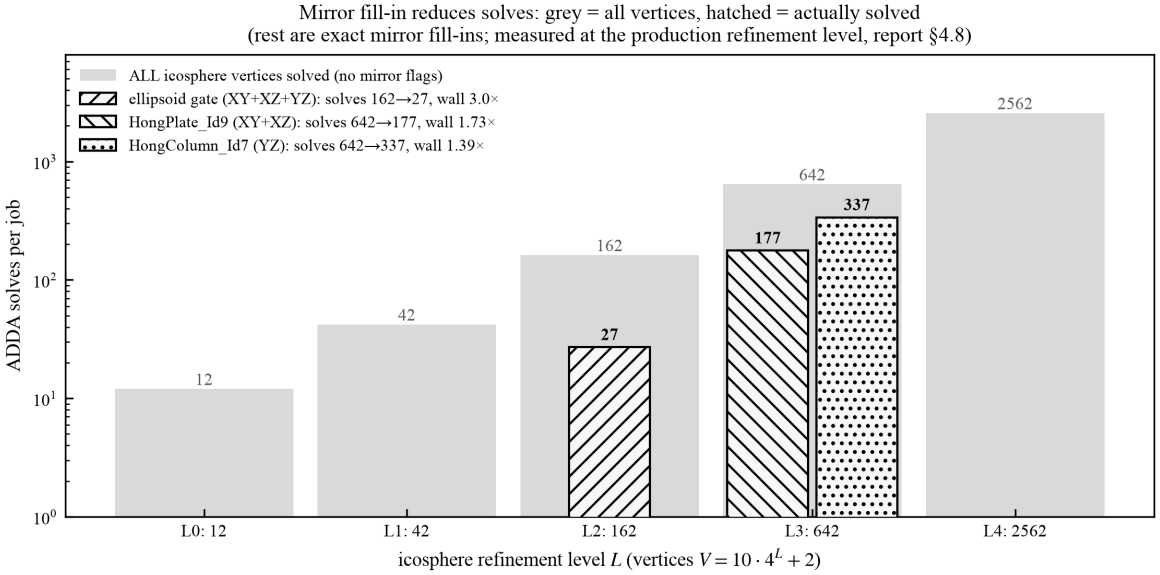


Fig. 5 — what mirror fill-in buys (measured). *Grey bars: every icosphere vertex solved (no flags). Hatched bars: actual ADDA solves with the habit's flags at that level — the rest of the vertices are exact*

mirror fill-ins (still SH-expanded + cached, hence the wall-time gain in the legend is smaller than the solve-count ratio).

Wall gain is below the solve-count ratio at probe size because the per-*vertex* work (SH expansion + cache write) still happens for fills; at production $x \approx 10$ the ADDA solve dominates and the ratio closes. `sym_Zobj` (C_n rotation) is parked: exact C_{2z} itself is resolution-dependent (HongPlate: 1.000 fine / 0.986 coarse), and the cube-lattice model never realizes ideal C_6 exactly; XY+XZ fill-in already implies exact C_{2x} anyway.

1.8 What the SSDB export contains

Post-processing turns the per-job store into ARTS-SSDB NetCDF — the format of the published ARTS Microwave SSP Database, whose TRO generation is documented in **Eriksson et al. (2018, ESSD 10, 1301–1326)**; the ARO extension of that format is documented in **Brath et al. (2020, AMT 13, 2309–2333)**. Our writer version is `beta_0.9.1` (Phase-6 additive backscatter variables; the 7 parallel variable lists all grow in lockstep). One file per **(β, size, f, T)** under a nested tree:

```
<adda_results>/SSD/Azimuthally_random/ice/Aggregate/Pristine/HongPlate_Id9/
    AzimuthallyRandom_beta090deg/
        Dmax00019um_Dveq00017um_Mass1.086e-11kg/
            Freq166.900GHz_T270K.nc
```

Physically meaningful content per file (all spectral, units m^2):

- `phaMat_data_real/imag` — phase-matrix spherical-harmonic coefficients, dims (16 Mueller elems × 1 az-inc × 19 za_inc × n_sph_coeffs);
- `extMat_data` — extinction-matrix elements (K11, K12, K33) on the (az, za_inc) grid;
- `absVec_data` — absorption vector (2 elems), computed as $C_{\text{ext}} - C_{\text{sca}}$ from the spectra, with a **negative-a1 rescale diagnostic** (§3.3);
- **`backMat_data` + `backMat_index`** (Phase 6) — the orientation-averaged exact-backscatter Mueller matrix, m²/sr, dims (phaMatElem × aa_inc × za_inc); `backMat_index` is the fixed identity pairing like `phaMat_index`. This is the *spectrally independent* backscatter value (not reconstructed from the phase matrix, §1.4 item 3); NaN-filled for stores that predate Phase 6;
- `ShapeData` scalars: diameters, mass, dpl, N_dipoles, refractive index (real/imag), plus provenance attrs (`refrIndex_model='Maetzler2006'`, habit id/name, source file);
- `CalculationData` attrs (provenance of the run): ADDA version, cluster, n_cores, `ADDA_eps`, `Refinement_delta`, `Mueller_sph_eps`, `Phase_sph_eps`, symmetry flags, shtns version.

The derived quantities consumers usually want (C_ext, SSA, asymmetry parameter g) are **not** stored — they are computed downstream by the published-DB tooling (`arts-sspdb-interface/`). Full variable table: Appendix C.

1.9 Analytic diffraction-peak model + residual storage (Phase 9A — scanned, not yet in the pipeline)

*Status: scanned and gated end-to-end (2026-11-09, `analysis/phase9_fwdmodel_scan.py`, plan `plans/forward_peak_plan.md`, report §9) — **no pipeline code changed**. This section develops the idea the way a textbook would: §1.9.1 fixes the amplitude language, §1.9.2 derives why a large particle must have a narrow forward peak, §1.9.3 turns the peak width into a spectral-order budget, §1.9.4 presents the model/residual split, §1.9.5 derives the model itself and its validity conditions, §1.9.6 reports the measurements, and §1.9.7 the verdict.*

1.9.1 The language: scattering amplitude, Mueller matrix, optical theorem

Everything below speaks in terms of the **complex far-field amplitude**. With $k = 2\pi/\lambda_0$ the scattered field of one solve is written, in the usual far-field notation,

$$\mathbf{E}_{\mathrm{sc}}(\mathbf{r}) \;=\; \frac{e^{ikr}}{r}\, f(\hat{n})\, \mathbf{E}^0_{\mathrm{inc}}\,, \qquad \hat{n} = \mathbf{r}/r, \tag{1.9.1}$$

where f is the 2×2 amplitude (Jones) matrix in the local scattering basis. This is exactly the object the pipeline already parses: the columns of ADDA's `ampl_scatgrid` are the Bohren–Huffman (BH) elements $s_1..s_4$ under the standard kf normalization (§1.4). The measured quantity is the **Mueller matrix**, a *quadratic* function of the amplitude,

$$M(\hat{n}) \;=\; f(\hat{n}) \otimes f^*(\hat{n}) \quad \text{(Stokes bilinear form)}, \tag{1.9.2}$$

which is what gets expanded in spherical harmonics and stored (§1.6). The third player is the **optical theorem**, which extracts extinction from the amplitude *linearly*:

$$C_{\mathrm{ext}} \;=\; \frac{4\pi}{k}\, \mathrm{Im}\big[\hat{s}^{\dagger} f(\hat{s})\, \hat{s}\big], \tag{1.9.3}$$

where $\hat{s}$ is the propagation direction of the incident beam — Eq. (1.9.3) is precisely the Mishchenko construction behind `acf.extinction_matrix` (§1.4). Note the contrast that will control everything in §1.9.4: **M is quadratic in f, C_{ext} is linear in f.**

Two limiting regimes of $f(\hat{n})$:

- **Rayleigh regime** ($x = \pi D_{veq}/\lambda_0 \ll 1$): the particle is a driven dipole; $|f|^2 \propto \cos^2\theta$ per polarization channel — smooth, **no spike** anywhere. (An early note in this manual claimed the forward peak "dominates at very small x" — wrong; the Phase-9A plan corrected the framing.)
- **Geometric regime** ($x \gg 1$): the pattern is dominated by a narrow **diffraction peak** around $\hat{n} = \hat{s}$. Production runs live at $x \le 10$, i.e. between the regimes — where the peak is real but only moderately narrow, and *exactly* what sets the angular sampling cost (Fig. 13a shows the transition for exact-Mie spheres: no spike at $x = 1$, a 4-decade spike at $x = 10$).

1.9.2 The extinction paradox and the diffraction peak (Babinet)

Classical result (van de Hulst; B&H): for $x \to \infty$ the extinction cross section of *any* particle tends to **twice its geometric projected area** G: $C_{\mathrm{ext}} \to 2G$. Pure ray tracing through the particle removes only G — the geometric shadow — so where does the other half come from? Babinet's principle answers: the shadow behind the particle must be dark, which requires a field *radiated by the particle silhouette* that is exactly $\mathbf{E}^0_{\mathrm{inc}}$ out of phase there — and any field that cancels the incident wave at $\hat{n} = \hat{s}$ must, by conservation, reappear as coherent scattering *into* the forward direction. For a silhouette (aperture) A with eikonal amplitude transmittance $e^{i\Delta(\rho)}$, $\Delta = k(m-1)\ell(\rho)$, the diffracted amplitude is, up to a sign convention fixed by the §1.9.5 anchor,

$$f_{\mathrm{diff}}(\hat{n}) \;=\; \frac{ik}{2\pi} \iint_A \big[1 - e^{i\Delta(\rho)}\big]\, e^{-ik\hat{n}\cdot\rho}\; d^2\rho\,. \tag{1.9.4}$$

Insert the black-screen limit ($1 - e^{i\Delta} = 1$) into the optical theorem (1.9.3): $f_{\mathrm{diff}}(\hat{s}) = ikA/2\pi$ gives $C_{\mathrm{ext,diff}} = (4\pi/k)(kA/2\pi) = 2A$ — a perfectly black screen of area A extinguishes $2A$: the paradox, and with it the statement that **the forward peak alone carries $C_{\mathrm{ext}} \to 2G$.**

Fourier duality fixes the peak's width. An aperture of projected radius $a_\perp$ radiates into a cone of half-angle

$$\sin\theta_1 \;=\; 1.22\,\frac{\lambda_0}{2a_\perp} \;=\; 0.61\,\frac{\lambda_0}{a_\perp} \quad \text{(first Airy zero)}, \tag{1.9.5}$$

so the peak narrows like λ_0/D: for a sphere ($a_\perp = D_{veq}/2$, $x = \pi D_{veq}/\lambda_0$) the first minimum sits at $\theta_1 \approx 220^\circ/x$ — verified in Fig. 13b (the dashed rings at 44° for $x = 5$, 22° for $x = 10$). The sidelobes beyond the peak are set by the *projected silhouette*, not by D_{veq} — for plate habits the silhouette changes strongly with orientation, which is why the peak structure is orientation-dependent.

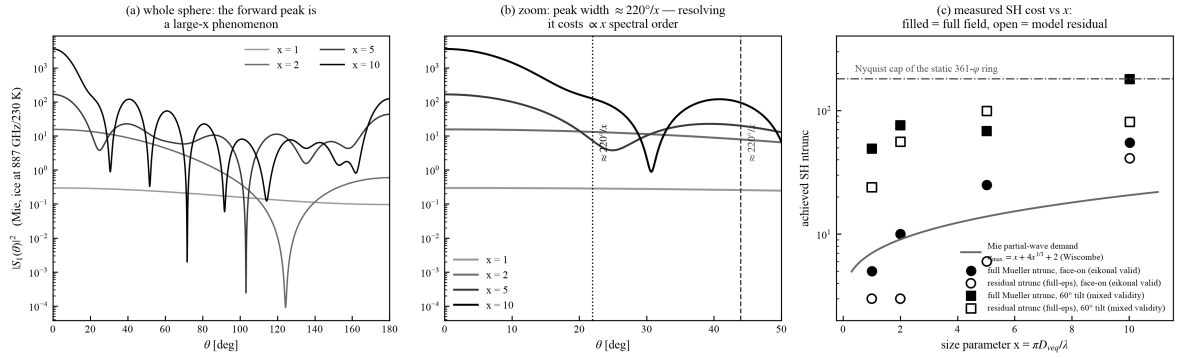


Fig. 13 — why the diffraction peak is the cost driver. *(a) exact-Mie $|S_1|^2$ over the whole sphere for four size parameters — the spike appears only at large x (greyscale ramp: darker = larger x); (b) zoom on 0–50°: the peak width shrinks like $220^\circ/x$; (c) achieved SH truncation order per production cell vs x: filled markers = today's full-field cost (grows like the grey Wiscombe curve and saturates at the dash-dotted Nyquist cap), open markers = what a model+residual split would need instead. Data:* `analysis/phase9_results.json`.

1.9.3 What the peak costs: the band-limit argument ($n_{\text{trunc}} \propto x$)

Spherical harmonics up to degree n form a **band-limited function space on the sphere**: they cannot represent angular structure finer than

$$\Delta\theta_{\min} \;\approx\; \frac{\pi}{n}\ \text{rad} \;=\; \frac{180^\circ}{n}\ \text{deg} \qquad \text{(equivalently a } 2n{+}1\text{-point } \varphi\text{-ring, the Phase-8 Nyquist rule §1.6.1).} \tag{1.9.6}$$

Therefore resolving a peak of width $\theta_1 \propto 1/x$ *requires*, for any numerical method,

$$n_{\text{trunc}} \;\gtrsim\; \frac{\pi}{\theta_1} \;\propto\; x\,. \tag{1.9.7}$$

The rigorous version of the same law for spheres is the Mie partial-wave demand — $n_{\max} = x + 4x^{1/3} + 2$ (Wiscombe; the $x^{1/3}$ term is the edge/"creeping-ray" boundary layer) — and it is the same law the Phase-8 grid budgets use (§1.6.1). The measured per-vertex ntrunc of the production pipeline follows it (Fig. 13c) and even **saturates the static grid's Nyquist cap** (ntrunc 179 ≈ 180 = the cap of the 361-point φ-ring) at $x = 10$, tilted orientations.

Why this is a *cost* statement: the SH coefficient count per vertex scales $\propto n_{\text{trunc}}^2$ (storage in `mueller_data.nc`, per-vertex caches), as do the shtns analysis/synthesis and the Wigner rotations of the orientation average (§1.5–1.6). **The diffraction peak is the single biggest data/cost item at large x — and (1.9.7) says numeric sampling of the peak can never be made cheaper. The only remaining lever is to stop storing the peak numerically.**

1.9.4 The book-keeping: model the peak analytically, store only the residual

Radiative-transfer has long "renormalized" the forward peak away (the δ-Eddington / δ-M trick: replace the peak by a delta function and preserve its lowest moments — at the cost of angular resolution *inside* the peak). The Phase-9 idea does the opposite: keep the peak **fully resolved**, but represent it by an analytic formula that is re-evaluated on demand, and store numerically only the remainder. Formally, split the **amplitude** field

$$f(\hat{n}) \;=\; F_{\text{model}}(\hat{n}) \;+\; g(\hat{n}) \tag{1.9.8}$$

with F_{model} analytic (§1.9.5) and $g \equiv f - F_{\text{model}}$ the residual. Only g needs a grid and SH storage. Three structural facts make this safe — or dangerous, if misapplied:

1. **The identity is exact by construction.** g is *defined* as $f - F_{\text{model}}$, so $F_{\text{model}} + g \equiv f$ regardless of the model's quality. The model's own error is never part of the error budget; it only determines *how smooth and small* g is. A bad model degrades gracefully ($g \approx f$, zero savings) — **unless** it makes g rougher or larger than f, which is exactly the failure mode measured in §1.9.6.
2. **The split must happen at amplitude level, not Mueller level.** By (1.9.2),

$$M[f] \;=\; M[F_{\text{model}}] \;+\; M[g] \;+\; 2\,\mathrm{Re}(F_{\text{model}} \otimes g^{*}). \tag{1.9.9}$$

The cross-term is $\mathcal{O}(|F||g|)$ — near the peak, where $|F|$ is largest, it is comparable to $M[F_{\text{model}}]$ itself. Storing "$M - M[F_{\text{model}}]$" numerically would therefore require the consumer to carry F anyway and would destroy the interference structure. Keeping (F_{model}, g) keeps everything algebraically exact. 3. **Extinction comes out of the same identity for free.** The optical theorem (1.9.3) is *linear* in f:

$$C_{\text{ext}}[f] \;=\; C_{\text{ext}}[F_{\text{model}}] \;+\; \frac{4\pi}{k}\,\mathrm{Im}[\hat{s}^{\dagger} g(\hat{s})\hat{s}], \tag{1.9.10}$$

and $F_{\text{model}}(\hat{s})$ is analytic — the forward-amplitude column of §1.4 survives the split.

The error rule follows immediately from fact 1: the reconstruction error is the SH truncation error of g, an **absolute** quantity, so the residual's truncation target must be measured against the *full field's* intensity scale,

$$\delta[g] \;\le\; \texttt{eps_sphexp} \cdot \mathrm{RMS}(M_{11}^{\text{full field}}) \qquad \text{not} \qquad \texttt{eps_sphexp} \cdot \mathrm{RMS}(M_{11}^{g})\,. \tag{1.9.11}$$

This "**full-eps**" rule is the single most consequential methodological finding of Phase 9A; the historic own-eps rule (§1.6) silently misprices residuals — a worked example in §1.9.6.

1.9.5 The model itself: eikonal ray-march + analytic radiation

The model must exist for *arbitrary voxel habits at arbitrary orientations* (exact Mie exists only for spheres), and it must be cheap and orientation-local. The construction has three steps.

Step 0 — an exact starting point. For a particle of constant refractive index m, the scattered far field is produced by the polarization current $\mathbf{P} = (m^2 - 1)\mathbf{E}$ *inside* the volume. Taking the far-field limit of the radiation integral gives an identity with **no approximation at all**, provided the true internal total field is inserted:

$$f(\hat{n}) \;=\; \frac{k^2}{4\pi}\,(m^2 - 1) \iiint_V \frac{E_{\text{tot}}(\mathbf{r})}{E_{\text{inc}}^{0}}\; e^{-ik\hat{n}\cdot\mathbf{r}}\; d^3r\,. \tag{1.9.12}$$

(Complex amplitude shorthand; applied per polarization channel like (1.9.1).) This is the standard "distorted-wave" representation: everything hard is hidden in E_{tot}.

Step 1 — the eikonal approximation to E_{tot}. Approximate the internal field by a plane wave that accumulates the optical path along each incident straight ray: with foot point ρ in the screen plane through the scatterer center perpendicular to the incidence direction $\hat{s}$, arc length s along the ray, and $\ell(\rho)$ the chord length through the *exact voxel set* on that ray,

$$\frac{E_{\text{tot}}}{E_{\text{inc}}^{0}} \;\approx\; \varphi(s) \;=\; e^{\,ik[s\,+\,(m-1)\,\ell(\rho)]}. \tag{1.9.13}$$

Single pass, straight rays: no refraction (bending), no internal reflection, no internal multiple scattering. Crucially, $\ell(\rho)$ is **ray-marched voxel-exactly from the shipped `.adda` file** — the geometry enters without any fitted ellipsoid or screen-plane shortcut, which is what makes the model habit-general.

Step 2 — carry the ray integral into (1.9.12). With $\mathbf{r} = \rho + s\hat{s}$ and $\mu = \hat{n}\cdot\hat{s} = \cos\theta$ the volume integral factorizes into a screen integral of a ray integral:

$$F_{\text{model}}(\hat{n}) \;=\; \frac{k^2}{4\pi}\iint e^{-ik\hat{n}\cdot\rho}\left[\int (m^2-1)\,\varphi(s)\,e^{-ik\mu s}\,ds\right] d^2\rho\,. \tag{1.9.14}$$

The bracket (the code's $G(\mu,\rho)$, evaluated chunked over screen pixels so the huge $(n_\mu \times n_\rho)$ matrix never materializes) is the *full-angle* form: replacing $\mu \to 0$ inside the bracket recovers textbook Fraunhofer diffraction at a phase-modulated screen (the eikonal transmittance $e^{i\Delta}$ of (1.9.4) reappears; for small phase delay ψ (below) the two agree up to constants absorbed into the calibration). Keeping μ is what makes F_{model} defined over the whole 4π grid rather than only a small cone, degrading gracefully away from the forward direction. In the project's kf-normalized ADDA convention the implementation evaluates the prefactor as $k^3/4\pi$ — i.e. $S_{\text{model}} = k\,F_{\text{model}}$ — so it can be compared and calibrated directly against the parsed `ampl_scatgrid` columns.

One global calibration constant c. Per run (T, f, size), a single complex number is least-squares-fitted in *amplitude space* (never log space — the model's zeros poison log fits) between $c\,F_{\text{model}}$ and the full ADDA grid. Physically c absorbs the local-field/depolarization correction that any distorted-wave ansatz misses: on a voxel sphere at $x = 0.3$ the fit returns $|c| = 0.614$, close to the exact uniform internal-field (Clausius–Mossotti) factor $3/(m^2+2) = 0.582$; at production x the eikonal limit is asymptotically right and the anchors give $c = 1.005 - 0.006j$ ($x = 1.9$) / $1.008 - 0.011j$ ($x = 3.8$) — a $|c-1| < 1\%$ agreement, cross-x drift 0.55%.

Why the conventions must be pinned first. The cross-term in (1.9.9) has amplitude $2|F||g|\cos(\text{phase slip})$: a phase slip of even a few mrad between the analytic model and ADDA's grid rotates part of "$-F$" into "$+F$" and turns the subtraction into an addition *precisely where the field is strongest*. The scan therefore anchors every convention on the one shape where truth is analytic (sphere, exact Mie — triple-validated scipy == mpmath == miepython == published Wiscombe MIEV0 case-14, $\le 5\cdot 10^{-16}$; the B&H 5.12 log-derivative D_n recurrence in `analysis/phase6_backscatter.py` is confirmed correct, gotcha §D-24). Anchor results (G1, all PASS): ADDA's `ampl_scatgrid` columns = BH $s_1..s_4$; ADDA's far-field phase origin = **scatterer center** (fitted offset $u_0 = 0.000$ lattice units — the earlier "bbox corner" hypothesis is dead); ADDA C_{ext} vs analytic at dpl 70 agrees to 0.23 % ($x = 1.9$) / 1.33 % ($x = 3.8$) — DDA discretization level; and a Mueller grid rebuilt from the parsed amplitudes matches ADDA's own `mueller_scatgrid` to $\le 7.4\cdot 10^{-11}$ (port check).

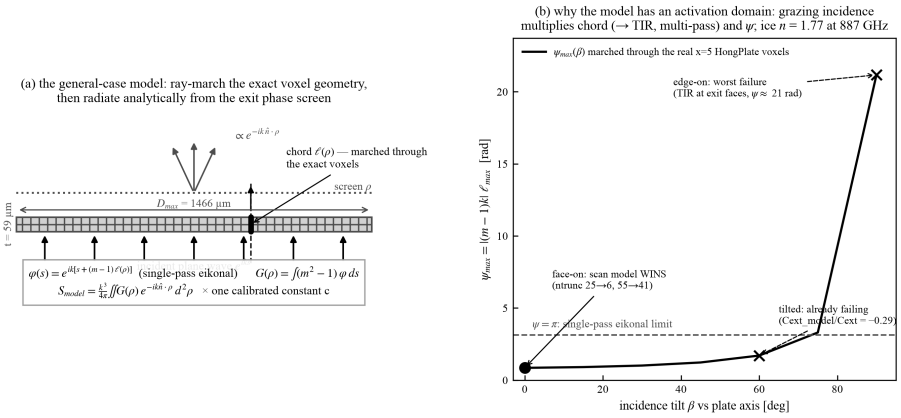


Fig. 14 — the model: what it is, and the one knob that says when it works. *(a) schematic on the real x=5 HongPlate slab: march every ray through the exact voxels to get the chord $\ell(\rho)$, radiate analytically from the exit screen; (b) the phase thickness ψ_{max} marched through the real voxels vs incidence tilt: below the dashed $\psi = \pi$ line the single-pass eikonal picture is plausible, far above it (edge-on) the model is guaranteed to fail. Ice refractive index at 887 GHz makes even a 541 µm plate "optically thick" edge-on.*

Validity conditions (textbook: van de Hulst / B&H, "diffraction approximation"). The single-pass eikonal (1.9.13) is justified only if

- (a) **refraction is weak**, $|m-1| \ll 1$ — *never* true for ice in the sub-mm ($n = 1.77$ at 887 GHz); for face-on plates refraction mainly perturbs chord geometry, and the scan shows empirically that the surviving error is tolerable there — but (a) failing is why the validity domain must be *measured*, not assumed;
- (b) **internal phase delay is modest**,

$$\psi(\rho) \;=\; \left|(m-1)\,k\,\ell(\rho)\right| \;\lesssim\; \pi, \tag{1.9.15}$$

because once $\psi > \pi$, internally reflected / totally internally reflected multi-pass fields reorganize the exit field with interference structure that (1.9.13) cannot represent *at any calibration*; - (c) the silhouette is well resolved — satisfied automatically (voxel-exact marching).

ψ_{max} is computed for free by the same ray-march. Measured on the real $x = 5$ HongPlate voxels at 887 GHz (Fig. 14b): $\psi_{max} \approx 0.7$ rad face-on, ≈ 2 rad at 60°, ≈ 21 rad edge-on — versus the π single-pass limit: valid face-on, hopeless edge-on.

1.9.6 Measured: the winning domain and the failure domain (15-cell scan)

Scan design (plan §9A): `analysis/phase9_fwdmodel_scan.py`, ADDA 1.4.0 on the production static 181 × 361 grid and the production command mirror (`-shape read`, `-eps 3`, `eq_rad` $= D/2$), Matzler ice at 230 K. Cells: HongPlate_Id9 at 887 GHz (the size grid hits $x = 1, 2, 5, 10$ exactly: $D_{veq} = 108/215/541/1078$ μm) × orientations {face-on, 60° tilt, edge-on} — the $x = 1$ edge-on cell is skipped (sub-wavelength slab, degenerate) — plus HongColumn_Id7 at $x = 2$ and $x \approx 4.9$ × {along-axis, 60°}: **15 cells**. Per cell: calibrate c; split (1.9.8); measure the residual fraction $\|g\|/\|f\|$ (amplitude L2); re-expand the residual with the *production* expansion rule (`expand_mueller_2_spharmonics`, `eps_sphexp=0.005`) under both the own-eps and the full-eps criterion (1.9.11); sanity-check $C_{ext,\mathrm{model}}/C_{ext,\mathrm{ADDA}}$. Full table: report §9.2 / `analysis/phase9_fwdmodel_table.txt`.

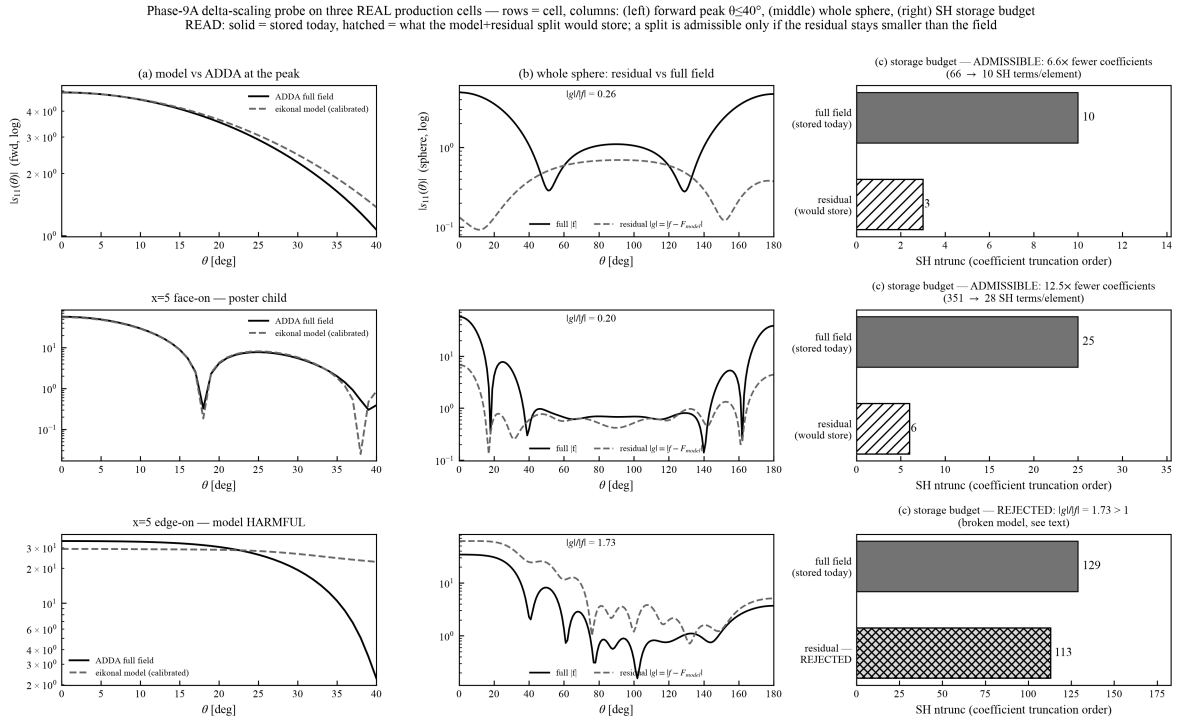


Fig. 15 — the three verdicts, per real production cell (row = cell). *Column 1: does the calibrated model reproduce the forward peak (black solid = ADDA, dashed = calibrated model)? Column 2: whole sphere — solid = full $|f|$, dashed = residual $|g|$; where the dashed curve rises above the solid one (edge-on row), the split makes storage WORSE. Column 3: the SH storage budget — solid bar = ntrunc stored today, hatched bar = ntrunc the model+residual split would store (full-eps rule, Eq. 1.9.11), cross-hatched bar = what-if bar REJECTED by the $|g|/|f|$ gate; the panel title carries the verdict with the coefficient-count ratio (n_{trunc} → SH terms = $(n+1)(n+2)/2$). The historic own-eps rule is* not* plotted — it is not decision-relevant (worked example in the metric rules below).*

Three archetypes emerge (Fig. 15):

- **WIN — HongPlate face-on at every x.** The model captures the peak shape *including the Airy zeros*: residual fraction $|g|/|f| = 0.03$–0.31, $C_{ext,\mathrm{model}}/C_{ext} = 1.05$–$1.15$. The full-eps residual ntrunc collapses: 5 → 3 ($x = 1$), 10 → 3 ($x = 2$), 25 → **6** ($x = 5$ — in the stored triangular $m \geq 0$ scheme the coefficient count per Mueller element is $(n+1)(n+2)/2$, so this is 351 → 28: a 12.5× smaller store), 55 → 41 ($x = 10$). Even the 60°-tilted $x = 10$ cell wins big: 179 → 81 — and the 179 there is the *grid-Nyquist-saturated* value (see the second rule below).
- **MARGINAL — small-angle deviations, tilted columns.** Intermediate cells (e.g. HongPlate $x = 2$–10 at 60°) shrink but less; the physics: ψ grows with tilt (Fig. 14b) and the single-pass picture frays.
- **HARMFUL — grazing incidence.** Plate edge-on at $x = 5$: $|g|/|f| = 1.73$ — the residual is LARGER than the field (possible because c is fitted over the full sphere; the model *adds* field the DDA does not have), and $C_{ext,\mathrm{model}} = -0.44\,C_{ext}$ — negative, the unambiguous signature of a broken model (§1.9.5 condition (b): $\psi \approx 21 \gg \pi$, internal TIR + multi-pass interference ignored). The 60°-tilted $x = 5$ cell fails the same way ($|g|/|f| = 1.21$, $C_{ext,\mathrm{model}} = -0.29\times$), and so does the column along-beam cell at $x \approx 4.9$ ($\psi \approx 8.6$ rad, $C_{ext,\mathrm{model}} = 0.16\times$, fitted $|c|$ collapses to ≈ 0.14 of the global value).

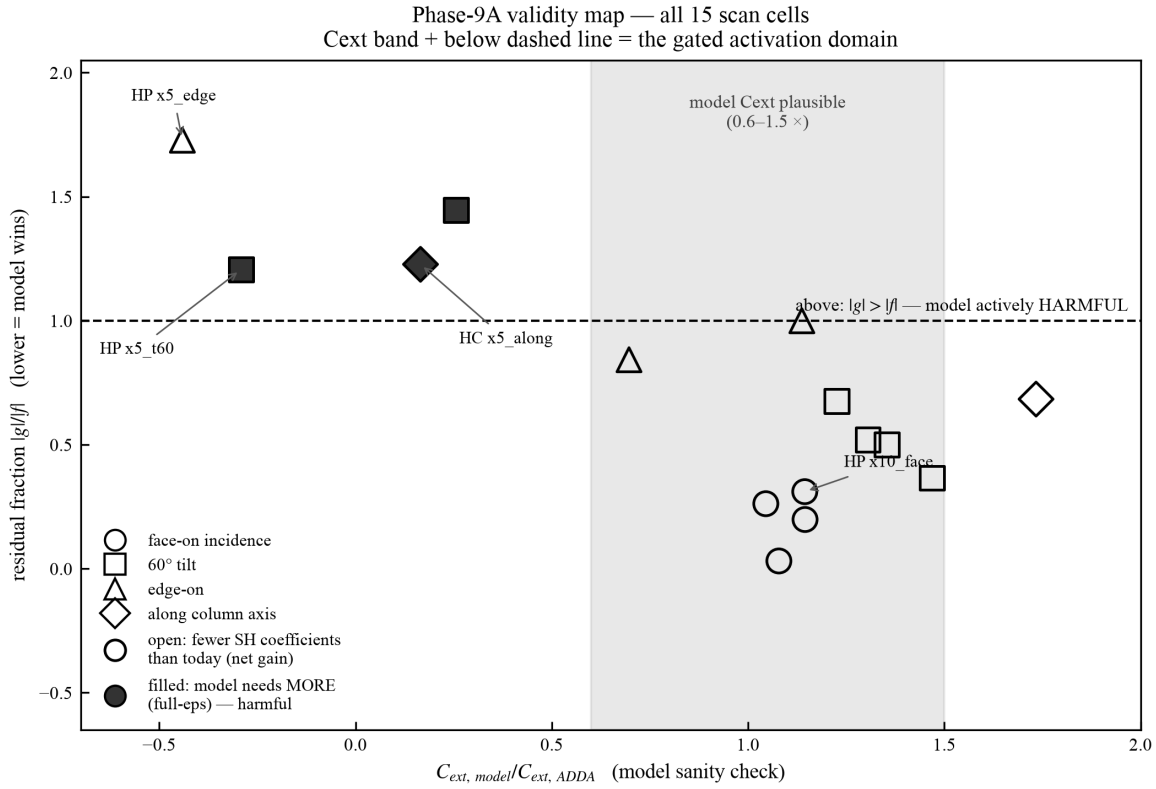


Fig. 16 — the gated activation domain for all 15 scan cells at a glance. *A cell is usable only if it sits in the shaded $C_{ext,model}$ band (sane, 0.6–1.5×) AND below the dashed line (residual smaller than the field). Marker shape = incidence orientation (legend bottom-left); marker fill = the storage verdict: open = the model+residual split needs FEWER SH coefficients than today, filled = it needs MORE (harmful). Every face-on cell is open and low, every tilted/edge-on cell drifts up and right — hence "gated", not blanket.*

Crucially, **validity is measurable without the full field**, i.e. cheaply per solve: the fitted calibration magnitude $|c_{\mathrm{fit}}/c_{\mathrm{global}}|$ deviates strongly exactly in the broken cells (dev > 1.4: $x = 2$ edge 5.2, $x = 5$ edge 6.4, $x = 10$ edge 4.0, HC $x = 5$ along 0.14), and $C_{ext,\mathrm{model}} \leq 0$ or $< 0.5\times$ flags them independently. Under the full-eps criterion, ntrunc is reduced in **10 of the 13 cells with** $x \geq 2$ (ratios 0.24–0.96); all three losses are validity-broken cells.

Two metric rules decide all of these numbers (and any future hybrid must respect them):

- **full-eps vs own-eps** (Eq. 1.9.11): judging the residual against its OWN small M_{11} norm misprices it badly — e.g. the $x = 1$ face cell: the residual's M_{11} norm is only 3% of the full field's, yet own-eps assigns the residual ntrunc **23**, worse than the full field's 5 (a 4.6× "penalty"), while full-eps reads **3** (an actual 1.7× saving — and 23/3 = a 7.7× overestimate of the residual cost). The rule of Eq. (1.9.11) is correct because the model is re-added exactly, so the residual's *absolute* truncation error rides on top of the full field for free;
- the $x = 10$ t60 **full** field already sits at the static grid's Nyquist cap (179≈180) — full-field twins of large-x runs are **grid-limited**, not expansion-limited (report §9.3), which bounds what any full-field comparison can show.

Backscatter: the model contributes nothing at $\theta = 180°$ (its field is off by $10^{-2} \ldots 8\times$, no consistent scale — the eikonal single-pass picture has no account of the coherent retro-reflection): the Phase-6 exact `backMat_data` route (§1.4, §1.8) stays mandatory and a hybrid would exempt that ring, as Brath et al. (2020) already flag the exact-backscatter direction as unreliable in the SH route anyway.

1.9.7 Verdict and Phase-9B design consequences

Conditional GO — a per-cell *gated* hybrid, not a blanket subtraction. The validity domain of §1.9.5–1.9.6 is real but orientation-dependent, and §1.9.6 showed it is detectable from cheap per-solve observables. Design sketch (report §9.4): per (vertex, β) decision from c-fit consistency, the $C_{ext,\mathrm{model}} > 0$ sanity band, and the ψ bound (1.9.15) — deciding model+residual vs plain full field; mixed runs store the per-cell flag in the nc metadata; `rawdata2average` reconstructs $f = F_{\mathrm{model}} + g$ on load.

Where the payoff lands, in the language of §1.9.3:

- **Internal cost (the target):** the per-vertex expansion/storage $\propto n_{\mathrm{trunc}}^2$ collapses in qualifying cells (e.g. the ntrunc 179→81 class, which dominates the orientation-expansion refinement at large x). The ADDA *solve* itself is untouched (the scat grid is far-field post-processing), and the grid laws of §1.6.1 could relax for qualifying vertices.
- **SSDB format: NO-GO for now** — with a clean textbook reason: shipping F_{model} as SH coefficients would require the model's OWN high-n_{trunc} tail, which *is* the expensive content (1.9.7): the saving exists only if the consumer receives the model *descriptor* (chord map + c) and evaluates (1.9.14) itself — a format change plus ARTS/RTTOV-side implementation; Phase-8.8 export caps already deliver 15× without any of that.
- Before building it, 9B must benchmark against a plain adaptive-eps re-balance (no model) — Phase-8's laws already capture part of the same trend.

Part 2 — Architecture & data flow

2.1 ARO call graph (production path)

Verified call graph (report §1.1, line numbers re-verified against current files):

Solve phase (one job per T, f, size) — see figure page below.

Post-processing phase (run later, per habit — see §3.3) — see figure page below.

2.2 TRO call graph

Gotchas specific to TRO (all verified): the mistral batch branch of `adda_call_randomorient` raises `NotImplementedError` (never was implemented — now loud instead of a swallowed NameError); there is **no completion-skip logic** (every listed index is re-run); `liste` **row 8 must carry the temperature** (9-row format) or the folder gets named with the flat index as `T####K` (a loud warning guards this legacy path); `deq` in the folder name receives the equivalent **radius** (legacy naming).

2.3 Artifact data flow (what lives where, what is transient)

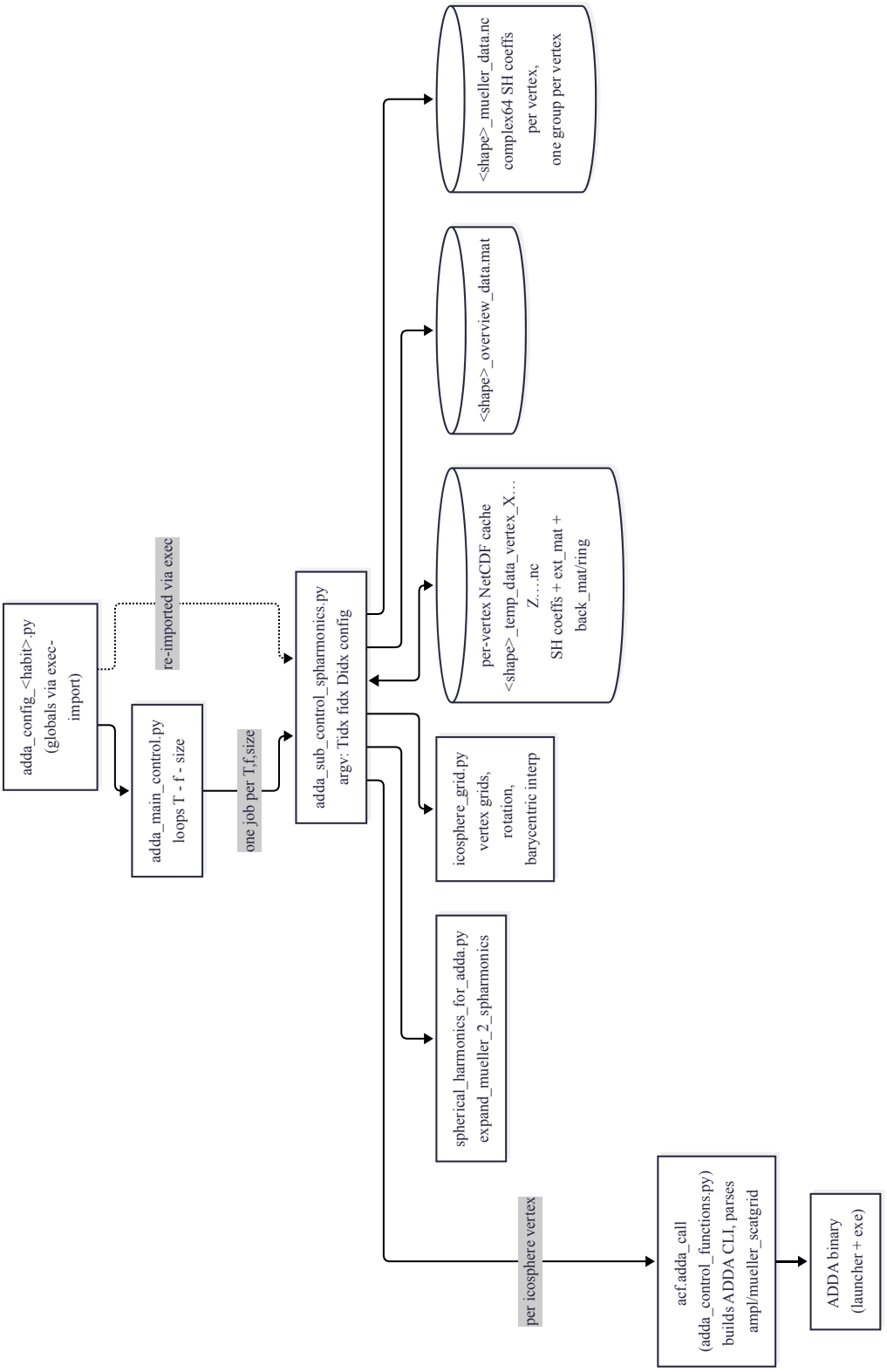
Idempotency layers (ARO): **(1)** job-level — main control skips a (T,f,size) if the folder contains at least one `*mueller_data.{xml,pkl,nc}` and `force_new=False`; **(2)** vertex-level NetCDF cache (survives a killed/restarted *job*, deleted only when the job completes); **(3)** within a refinement pass, vertices identical to the previous icosphere level reuse cached expansions without touching ADDA. TRO has **none** — plan reruns accordingly.

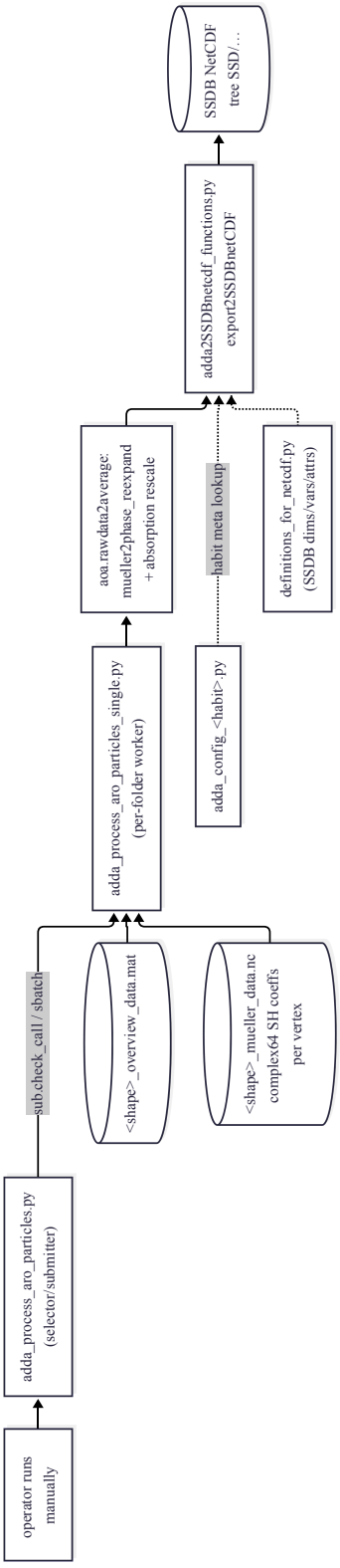
The final per-run Mueller store `<shape>_mueller_data.nc` (Phase 7) is one file per (T, f, size): one group per *computed* icosphere vertex (name = temp-cache `vertex_X…` id), holding the complex64 SH coefficients (`mueller_sph`, stored losslessly as float32 pairs), `relerror`, `ext_mat`, `back_mat`, and the explicit `sph_l`/`sph_m` coefficient-index arrays; the group attr `vertex_index` **is the sole ordering key** (never trust group iteration order). Written atomically (`.tmp` + `os.replace`). Legacy `.pkl` stores are still readable (deprecation warning; migrate with `analysis/migrate_pkl_to_nc.py`); legacy `.xml` is not readable anymore. Vertex-cache stores that predate Phase 6 lack `back_mat` — both cache-check sites then treat them as a **cache miss** and re-run ADDA, so a resumed old store can never silently yield NaN back data (full schema: report §7.1).

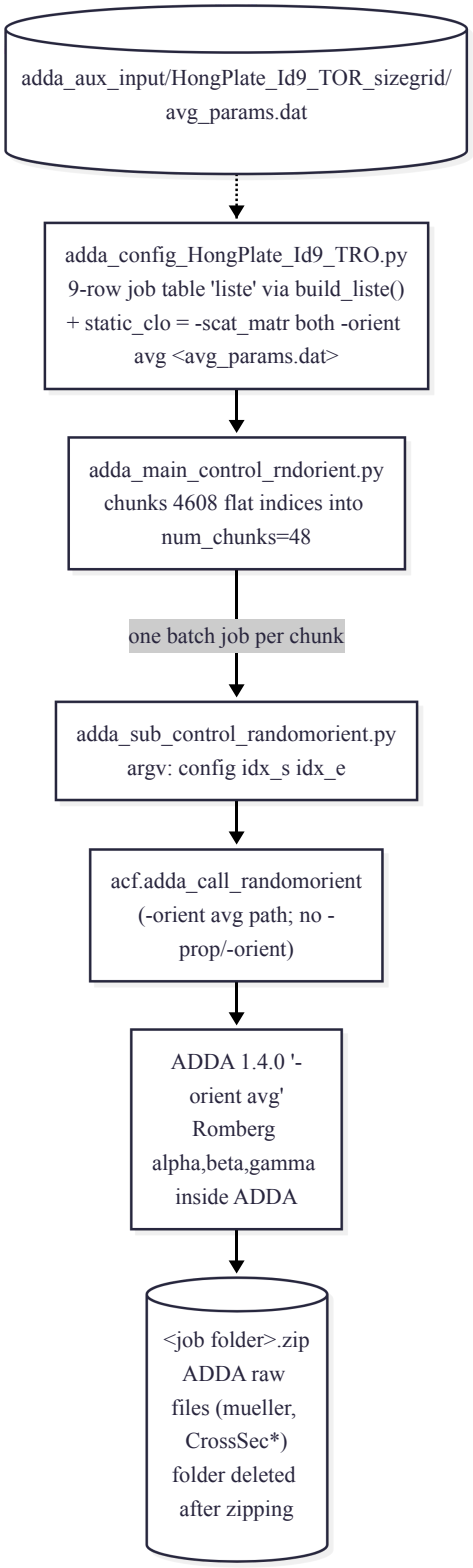
2.4 Module status table

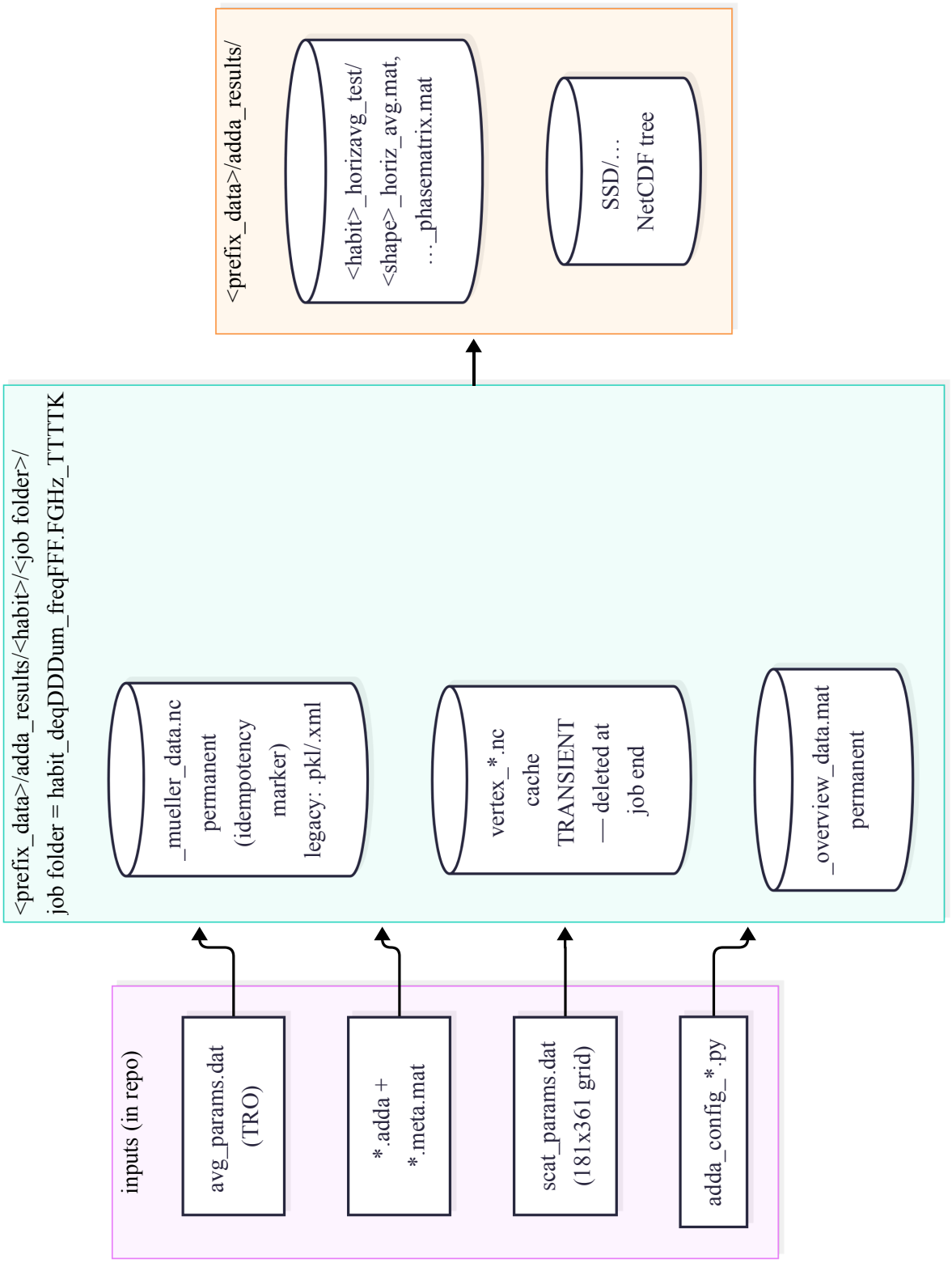
module	role	status
`adda_main_control.py`, `adda_sub_control_spharmonics.py`	ARO driver + per-job solver	**active**
`adda_main_control_rndorient.py`, `adda_sub_control_randomorient.py`, `acf.adda_call_randomorient`	TRO driver (`-orient avg`)	**active** (Phase-3 repaired)
`adda_control_functions.py` (acf)	ADDA CLI wrapper, Matzler dielectric, Romberg helpers, mirror transforms, NetCDF cache, MATLAB reader	**active**
`spherical_harmonics_for_adda.py` (spha), `icosphere_grid.py` (ico)	SH machinery, orientation grids	**active**

module	role	status
`adda_orientational_avg_spharmonics_functions.py` (aoa)	post-processing averaging / phase-matrix re-expansion	**active**
`adda_process_aro_particles[.py/_single.py]`, `adda2SSDBnetcdf_functions.py`, `definitions_for_netcdf.py`	post-processing + SSDB export	**active** (export gated behind env `SSDB_EXPORT_NETCDF=1`)
`adda_sub_control_gridref.py`, `acf.romberg_alpha/gamma`, `adda_orientational_avg_spharmonics_for_testing.py`	old grid-refinement experiment	**deprecated — do not extend**
`posix_spawn.py`	fast process spawn (Linux ctypes; macOS subprocess branch)	active helper
`memory.py`, `sbatch_adda`	legacy helpers	dormant
`old_configs/`, `SSD_scripts_deprecated/`, `NetCDF_export_testscript*.py`	archives / one-off test scripts	**archive — never import**


adda_config_ <habit> .py
(globals via exec-import)
re-imported via exec
adda_main_control.py
loops T - f - size
one job per T,f,size
adda_sub_control_spharmonics.py
argv: Tidx fidx Didx config
<shape>_mueller_data.nc
complex64 SH coeffs
per vertex,
one group per vertex
<shape>_overview_data.mat
per-vertex NetCDF cache
<shape>_temp_data_vertex_X…
Z….nc
SH coeffs + ext_mat +
back_mat/ring
icosphere_grid.py
vertex grids,
rotation,
barycentric interp
spherical_harmonics_for_adda.py
expand_mueller_2_spharmonics
per icosphere vertex
acf.adda_call
(adda_control_functions.py)
builds ADDA CLI, parses
ampl/mueller_scatgrid
ADDA binary
(launcher + exe)


operator runs manually
adda_process_aro_particles.py (selector/submitter)
sub.check_cell / sbatch
<shape>_overview_data.mat
<shape>_mueller_data.nc complex64 SH coeffs per vertex
adda_process_aro_particles_single.py (per-folder worker)
adda_config_<habit>.py
habit meta lookup
aoa.raw/data2average: mueller2phase_reexpand + absorption rescale
definitions_for_netcdf.py (SSDB dims/vars/attrs)
adda2SSDBnetcdf_functions.py export2SSDBnetCDF
SSDB NetCDF tree SSD/…


adda_aux_input/HongPlate_Id9_TOR_sizegrid/avg_params.dat
adda_config_HongPlate_Id9_TRO.py
9-row job table 'liste' via build_liste()
+ static_clo = -scat_matr both -orient avg <avg_params.dat>
adda_main_control_rndorient.py
chunks 4608 flat indices into num_chunks=48
one batch job per chunk
adda_sub_control_randomorient.py
argv: config idx_s idx_e
acf.adda_call_randomorient
(-orient avg path; no -prop/-orient)
ADDA 1.4.0 '-orient avg'
Romberg alpha,beta,gamma inside ADDA
<job folder>.zip
ADDA raw files (mueller, CrossSec*)
folder deleted after zipping


inputs (in repo)
avg_params.dat (TRO)
*.adda + *.meta.mat
scat_params.dat (181x361 grid)
adda_config_*.py
<prefix_data>/adda_results/<habit>/<job folder>/
job folder = habit_deqDDDum_freqFFF.FGHz_TTTTK
_mueller_data.nc
permanent
(idempotency marker)
legacy: .pkl/.xml
vertex_*.nc
cache
TRANSIENT
— deleted at job end
_overview_data.mat
permanent
<prefix_data>/adda_results/
<habit>_horizavg_test/
<shape>_horiz_avg.mat,
…_phasematrix.mat
SSD/…
NetCDF tree

Part 3 — Operations & runbooks

3.1 The config contract

Every habit has one `adda_config_<habit>.py`; the control scripts inject it with `exec('from adda_config_X import *')` and read its globals. **Keep field names** when copying a config — the drivers `getattr`-probe them. The canonical reference is `adda_config_HongPlate_Id9.py`. Annotated field → consumer map:

config global	meaning	consumed by
`calc_type` (`'mistral'`/`'thunder'`/`'local'`)	host mode; `local` = **Phase-3 no-queue macOS branch** (repo-relative paths, direct `sub.check_call`)	both main controls, sub-controls, export provenance
`prefix`, `prefix_data`, `main_path`, `ramdrive`	storage roots (local: repo-relative; `ramdrive=None` → system temp)	sub-controls (job folders, ADDA `-dir`)
`adda_prefix`, `cluster_name`	ADDA checkout root; SSDB provenance string	export `CalculationData`
`adda_binary`, `adda_launcher`, `adda_version`	ADDA executable triple; env overrides `ADDA_BINARY`/`ADDA_VERSION` apply on top; sub-controls fall back to 1.2 defaults if a config lacks these	both sub-controls via `main_command = launcher + [binary]`
`shape_name`, `shape_foldername`, `shape_path`, `subpath`, `shapelist`, `shape_str`	habit + geometry file list (`shape_str=[]` ⇒ `-shape read <file>` is prepended)	main/sub controls, export meta lookup
`Sizes`, `Dmax`, `fmin` (arrays parallel to `.mat` files)	physical size grid + resolution classes (read from `*.meta.mat` with `acf.loadmat`)	job loops + selection rule
`frequencies` (Hz), `temperatures` (K)	the production channel set (35 channels 1–886 GHz × 3 T for HongPlate)	job loop; dielectric evaluation
`size_parameter_limit` (=10)	x-cap	main control skip
`number_of_betas` (=5), `incidence_angles` (10 vals)	ARO **output** coordinate resolution during solve-averaging (post-processing re-grids to 19 × 19)	sub-control averaging stage
`delta` (=0.01), `limit_sph` (=4)	icosphere orientation convergence / max level	sub-control refinement loop
`eps` (=3)	ADDA solver residual exponent	`acf.adda_call -eps`
`eps_sphexp` (=0.005)	per-vertex Mueller SH truncation target	`spha.expand_mueller_2_spharmonics`
`sym_XYobj`, `sym_XZobj`, `sym_YZobj`	**mirror fill-in flags — must be set from the coarsest resolution class** (§1.7, `analysis/phase4_symmetry.py`)	sub-control skip/fill logic; export provenance
`alpha_0/beta_0/gamma_0` (0/0/0)	base particle attitude handed to `-orient`	sub-control
`force_new` (False)	recompute even if completed-marker exists	main control idempotency
`ZIPPING` (False)	keep per-vertex ADDA output in zips (auto-deleted if False)	`acf.adda_call`

config global	meaning	consumed by
`scat_param_path`, `static_clo`	ADDA static flags: `-store_scat_grid -scat_grid_inp <scat_params.dat> -scat_matr both` (`-Csca`/`-alldir_inp` removed — never parsed)	`acf.adda_call`
TRO only: `liste` (9 rows!), `num_chunks`, `batch_time`, `avg_param_path`	flat job table (rows: shape, **eq radius m**, eps, m_re, m_im, λ m, shape file, batch time, **T K**) + chunking + avg file	rndorient pair
HPC: `ncores`, `Partition`	Slurm sizing	sbatch wrappers

3.2 ARO job flowcharts

Driver: `adda_main_control.py`

Part 1 — startup and the nested loops — see figure page below.

Part 2 — resolution-class selection, idempotency check, dispatch — see figure page below.

> **(!)** local mode runs thousands of jobs **synchronously in one foreground process**. For anything beyond smoke tests, deselect sizes/frequencies in a probe config or run the vertex loop in parallel (see §3.6).

One job: `adda_sub_control_spharmonics.py` (per T, f, size)

Part 1 — init and the icosphere vertex loop, per refinement level $L = 0\ldots$ (min 3 passes, max `limit_sph` = 4) — see figure page below.

Part 2 — orientation average, refinement loop, outputs — see figure page below.

The mirror-fill transform (φ remap + Stokes signs + SH re-expansion) is sketched in §1.7; the same vertex grid also answers `interpolation_weights_icosahedric_grid` queries during the averaging stage — barycentric interpolation, `aoa.pol_changer` for the Stokes polarization basis.

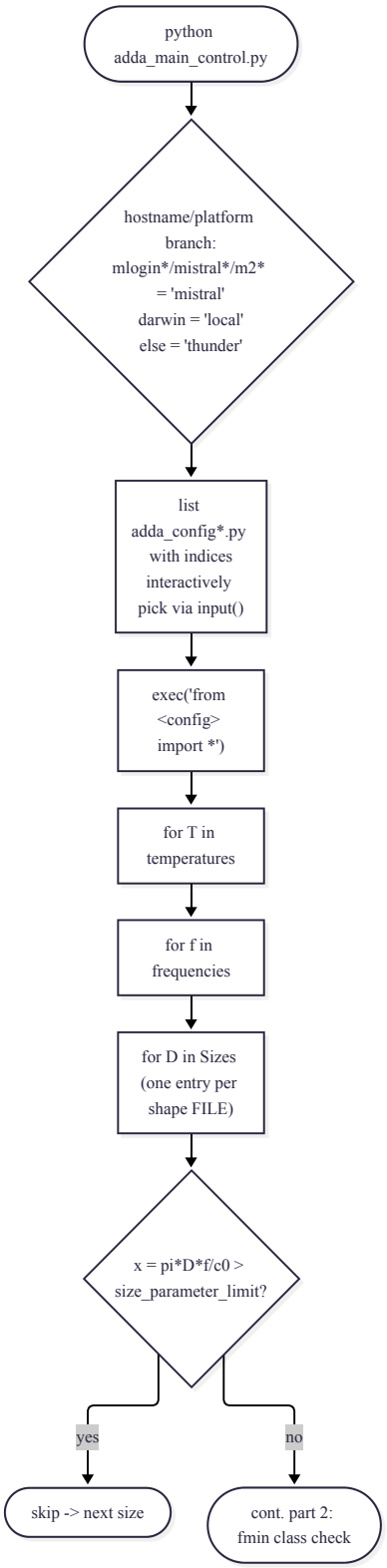
3.3 Post-processing: folder → SSDB NetCDF

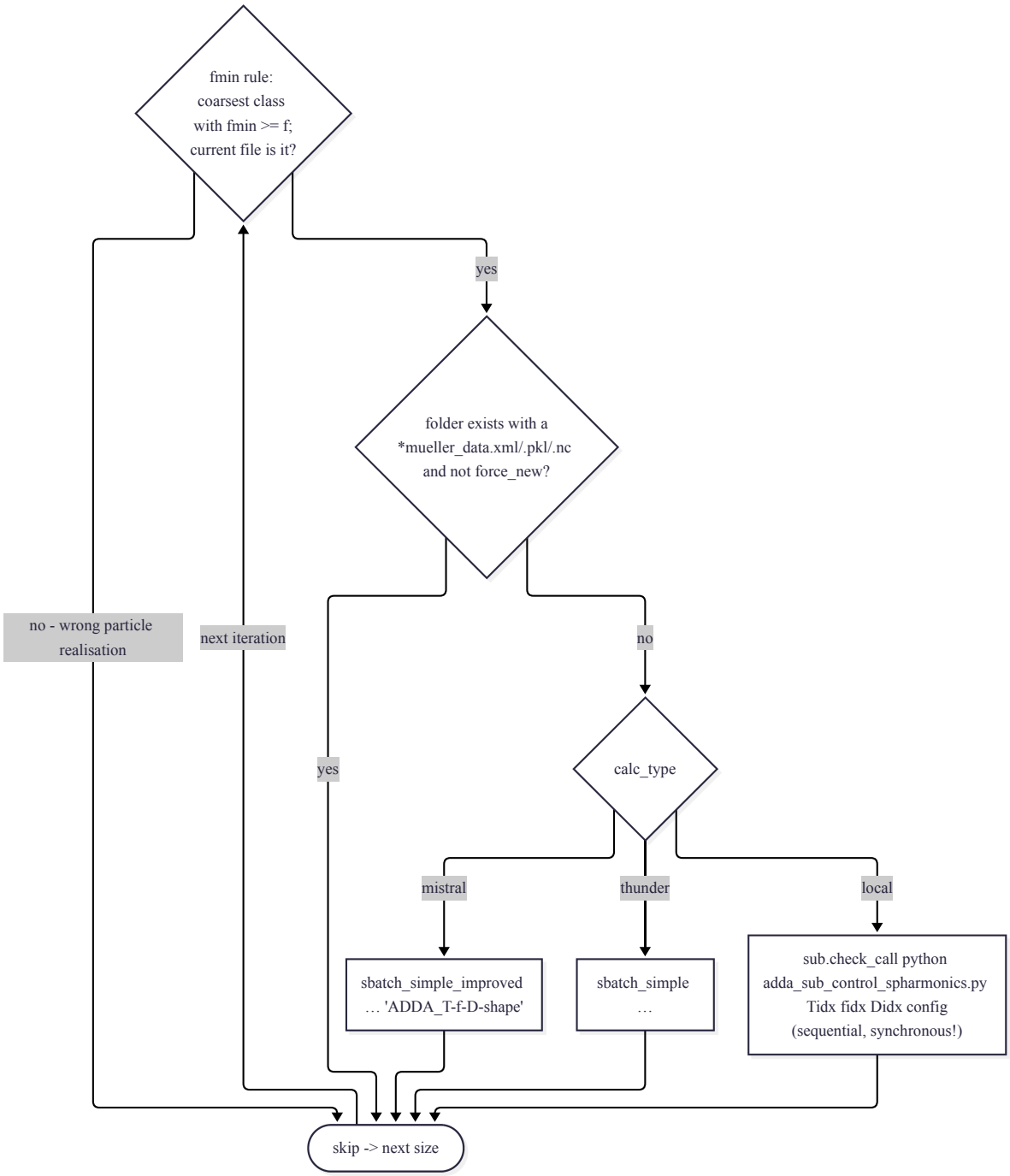
Run by hand per habit (or per folder) — this is deliberately decoupled from the solves — see figure page below.

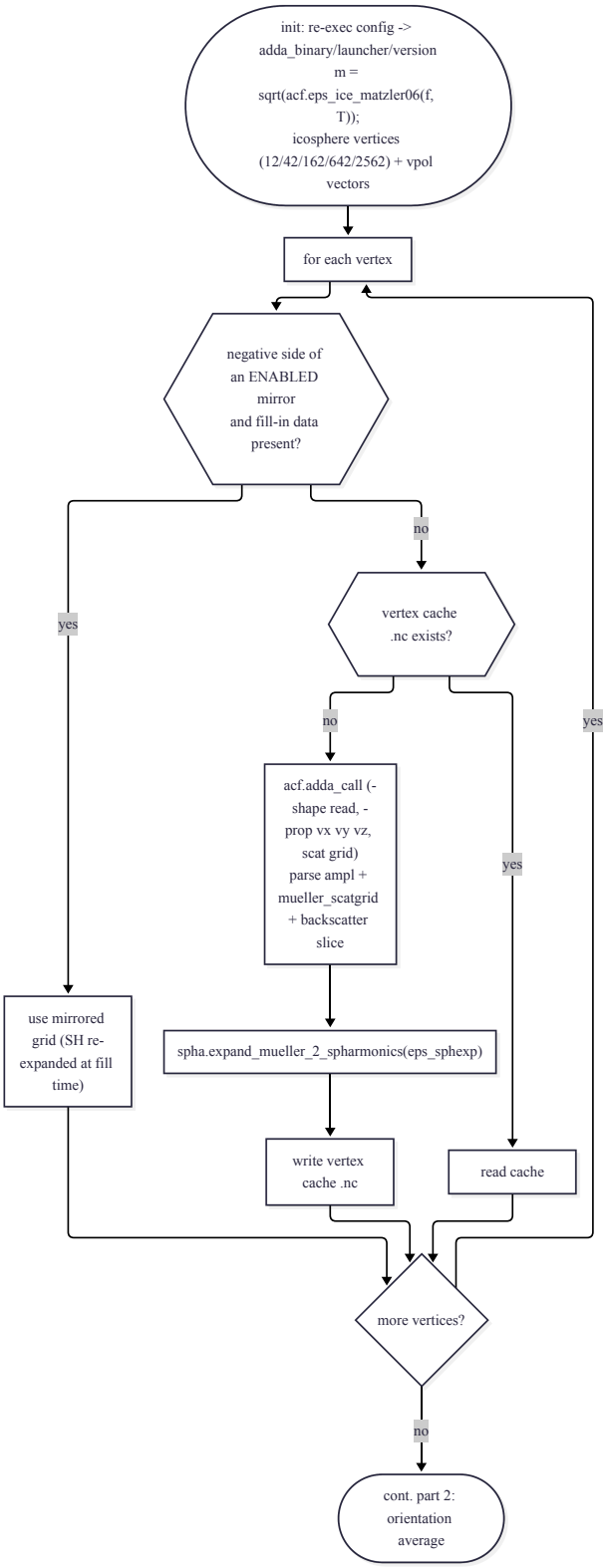
Notes: the 19×19 (β, θ_{inc}) grid and `eps_phase=0.001` are **hard-coded** in `adda_process_aro_particles.py:74-107` (independent of the solve-time 5×10); the SSDB export is **off by default** (`SSDB_EXPORT_NETCDF=1` to enable); the habit processed is env-selected `SSDB_HABIT=<habit>` (default `HexPlaAggCrystal_Id20`). The negative-a1 rescale is a physical-consistency repair: where the spectral C_sca exceeds C_ext the whole phase matrix at that node is scaled down; check the printed diagnostics — a large `frac_of_neg_a1` means the truncation error `eps_sphexp/eps_phase` is fighting you.

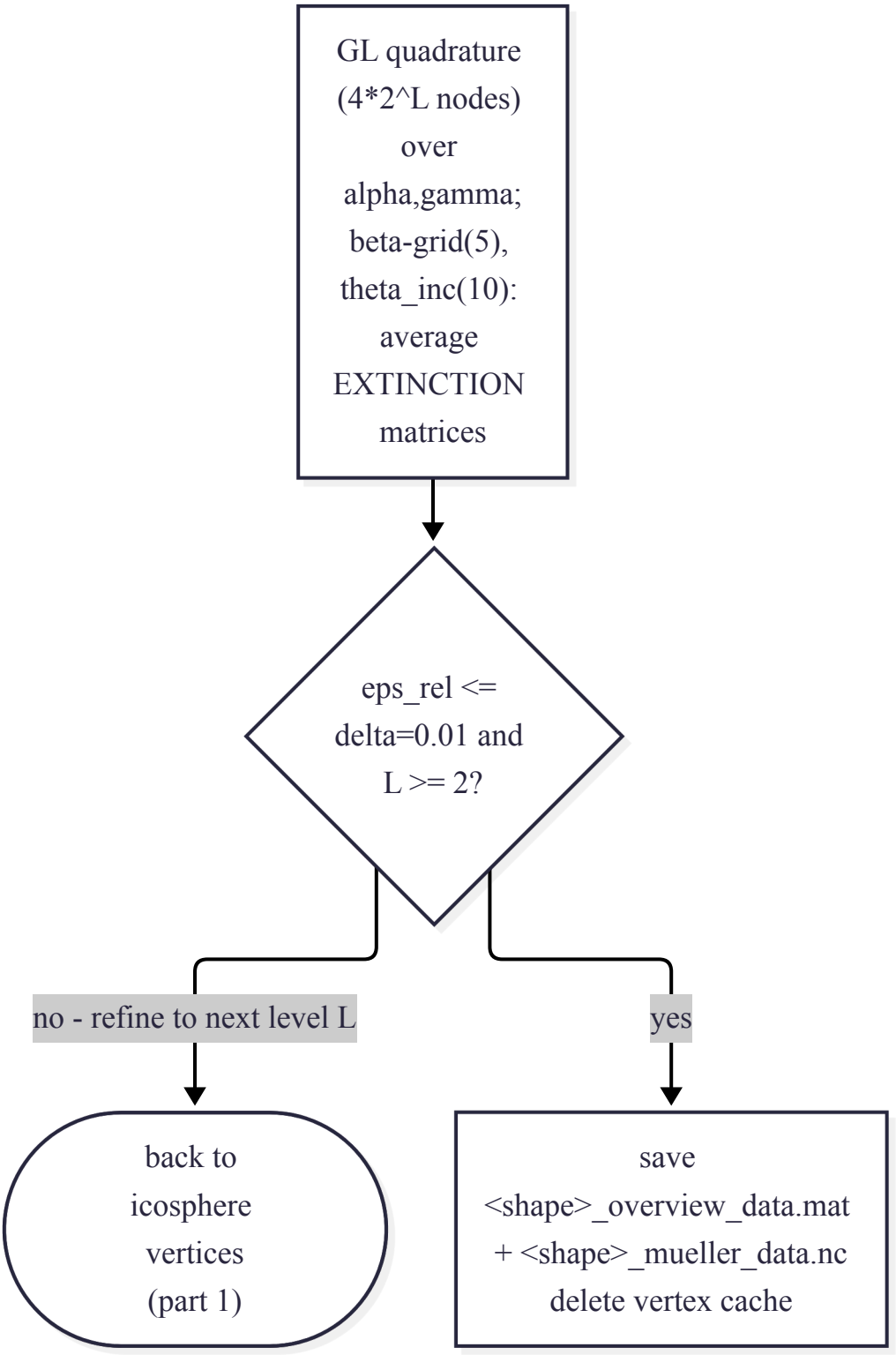
3.4 TRO production run

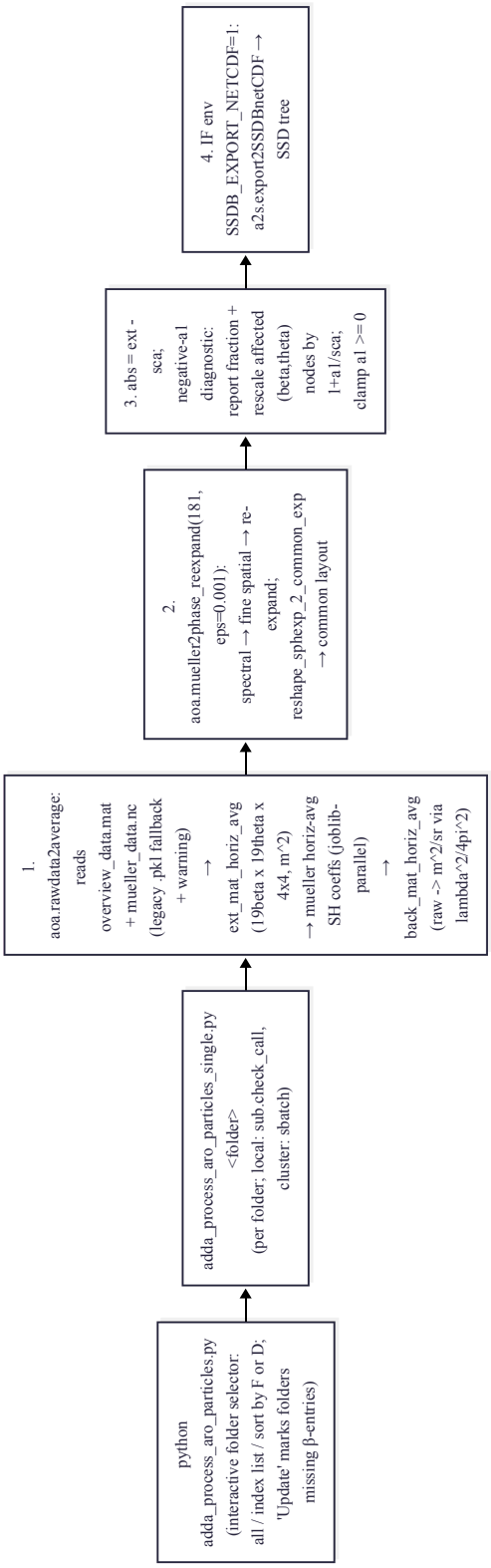
Gate artifacts kept in `analysis/`: `phase5_tro_beta_gate.py/.log` (β-fold equivalence), `phase5_tro_probe.log` (ADDA accepts the avg files in 1.2/1.4/1.5).

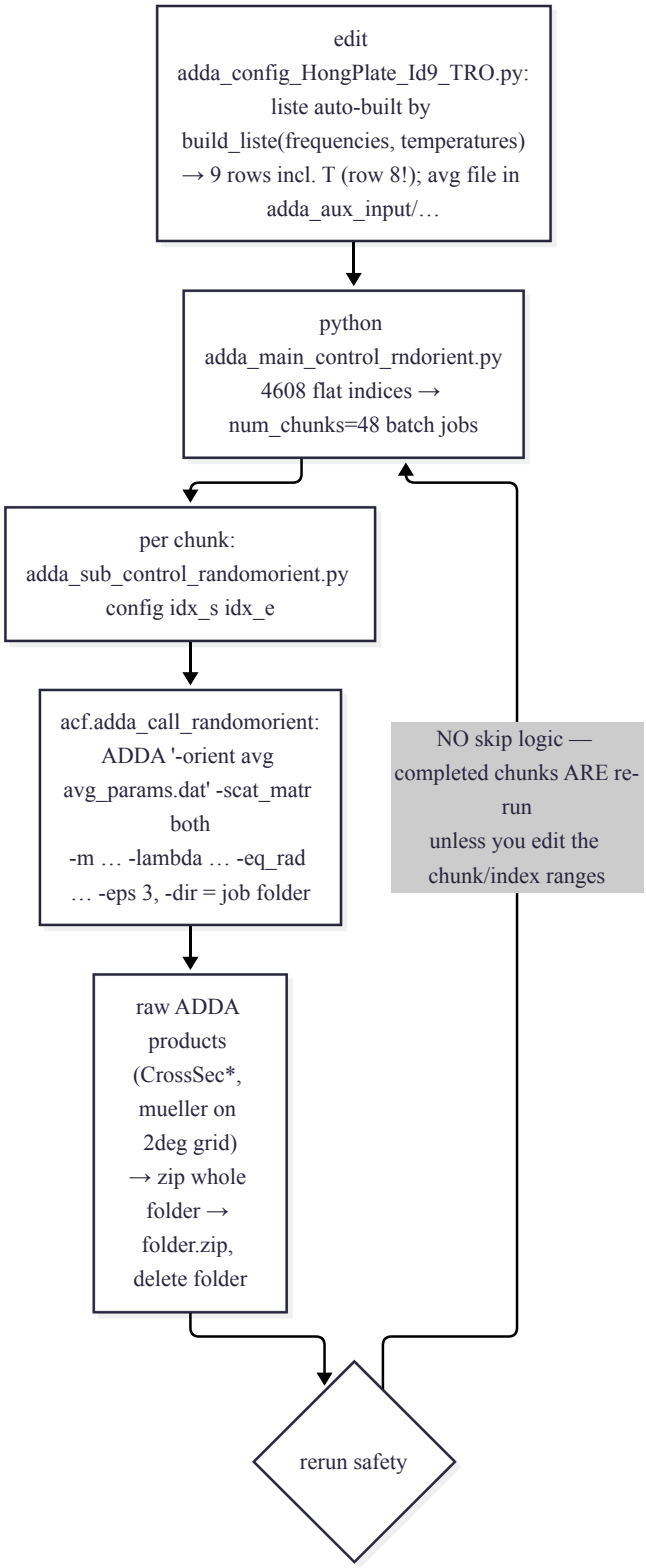

python
adda_main_control.py
hostname/platform
branch:
mlogin*/mistral*/m2*
= 'mistral'
darwin = 'local'
else = 'thunder'
list
adda_config*.py
with indices
interactively
pick via input()
exec('from
<config>
import *')
for T in
temperatures
for f in
frequencies
for D in Sizes
(one entry per
shape FILE)
x = pi*D*f/c0 >
size_parameter_limit?
yes
no
skip -> next size
cont. part 2:
fmin class check


fmin rule:
coarsest class
with fmin >= f;
current file is it?
yes
folder exists with a
*mueller_data.xml/.pkl/.nc
and not force_new?
no - wrong particle
realisation
next iteration
yes
no
calc_type
mistral
thunder
local
sbatch_simple_improved
… 'ADDA_T-f-D-shape'
sbatch_simple
…
sub.check_call python
adda_sub_control_spharmonics.py
Tidx fidx Didx config
(sequential, synchronous!)
skip -> next size


init: re-exec config -> adda_binary/launcher/version
m = sqrt(acf.eps_ice_matzler06(f, T));
icosphere vertices (12/42/162/642/2562) + vpol vectors
for each vertex
negative side of an ENABLED mirror and fill-in data present?
yes
no
vertex cache .nc exists?
yes
no
yes
use mirrored grid (SH re-expanded at fill time)
acf.adda_call (-shape read, -prop vx vy vz, scat grid) parse ampl + mueller_scatgrid + backscatter slice
spha.expand_mueller_2_spharmonics(eps_sphexp)
write vertex cache .nc
read cache
more vertices?
no
cont. part 2: orientation average


GL quadrature (4*2^L nodes) over alpha,gamma; beta-grid(5), theta_inc(10): average EXTINCTION matrices
eps_rel <= delta=0.01 and L >= 2?
no - refine to next level L
yes
back to icosphere vertices (part 1)
save <shape>_overview_data.mat + <shape>_mueller_data.nc
delete vertex cache


python
adda_process_aro_particles.py
(interactive folder selector:
all / index list / sort by F or D;
'Update' marks folders
missing β-entries)
adda_process_aro_particles_single.py
<folder>
(per folder; local: sub.check_call,
cluster: sbatch)
1.
aoa.rawdata2average:
reads
overview_data.mat
+ mueller_data.nc
(legacy .pkl fallback
+ warning)
→
ext_mat_horiz_avg
(19beta x 19theta x
4x4, m^2)
→ mueller horiz-avg
SH coeffs (joblib-
parallel)
→
back_mat_horiz_avg
(raw -> m^2/sr via
lambda^2/4pi^2)
2.
aoa.mueller2phase_reexpand(181,
eps=0.001):
spectral → fine spatial → re-
expand;
reshape_sphexp_2_common_exp
→ common layout
3. abs = ext -
sca;
negative-a1
diagnostic:
report fraction +
rescale affected
(beta,theta)
nodes by
1+a1/sca;
clamp a1 >= 0
4. IF env
SSDB_EXPORT_NETCDF=1:
a2s.export2SSDBnetCDF →
SSD tree


edit adda_config_HongPlate_Id9_TRO.py: liste auto-built by build_liste(frequencies, temperatures) → 9 rows incl. T (row 8!); avg file in adda_aux_input/…
python adda_main_control_rndorient.py 4608 flat indices → num_chunks=48 batch jobs
per chunk: adda_sub_control_randomorient.py config idx_s idx_e
acf.adda_call_randomorient: ADDA '-orient avg avg_params.dat' -scat_matr both -m … -lambda … -eq_rad … -eps 3, -dir = job folder
raw ADDA products (CrossSec*, mueller on 2deg grid) → zip whole folder → folder.zip, delete folder
NO skip logic — completed chunks ARE re-run unless you edit the chunk/index ranges
rerun safety

3.5 Local (macOS) runbook

Python env — mamba env `DDA` (py3.12): conda-forge `numpy scipy netcdf4 joblib matplotlib fftw pkg-config`; pip `typhon==0.10.0 shtns` (shtns: no osx-arm64 conda build; native wheel — rebuild with `pip install --no-build-isolation --no-deps shtns` while fftw+pkg-config are present). Never downgrade typhon (0.8.0 is broken on 3.12; `ty.arts.xml` is gone — Mueller artifacts are NetCDF4 `*_mueller_data.nc`, Phase 7).

Solver trees first — they are NOT in this repo. The ADDA trees are gitignored and were purged from the repository history on 2026-11-10 (report §11): a fresh clone contains no solver at all. Provision them as siblings of the repo root, keeping these folder names (configs resolve the binary as `adda_prefix + adda_binary`):

folder	ADDA version	how to obtain	role
`adda_1.4/`	**1.4.0**	`curl -L -o v1.4.0.tgz https://github.com/adda-team/adda/archive/refs/tags/v1.4.0.tar.gz && tar xzf v1.4.0.tgz && mv adda-1.4.0 adda_1.4`	**production** binary
`adda_1.2/`	1.2	same with `v1.2`	golden-baseline lineage
`adda/`	1.5.0-alpha3	`git clone https://github.com/m-brath/adda.git adda` (upstream has no 1.5 tag — its tags stop at v1.4.0)	cross-version checks
`adda-gui/`	—	`git clone https://github.com/m-brath/adda-gui.git`	reference only

Validated checkout states (pinned in `agents.md`): `adda` = `8f550a7`, `adda-gui` = `af66662`.

ADDA build (works identically for `adda_1.2`, `adda_1.4`, `adda/`; never run a stale Linux ELF binary that an older tree copy may still carry under `adda_1.2/src/`):

```
cd adda_1.4/src && make cleanseq                # stale Linux .d files break make otherwise
make seq "OPTIONS=NO_FORTRAN" \
  FFTW3_INC_PATH=$CONDA_PREFIX/../envs/DDA/include \
  FFTW3_LIB_PATH=$CONDA_PREFIX/../envs/DDA/lib
# make links WITHOUT rpath → relink with rpath so dyld finds libfftw3.3.dylib:
cd seq && gcc -o adda *.o -L<env>/lib -Wl,-rpath,<env>/lib -lm -lfftw3
```

(NO_FORTRAN is fine: only IGT/surface modes need Fortran; the pipeline uses `-int defint` default.)

Running a smoke job locally:

```
mamba activate DDA
cd adda_python_scripts
python adda_main_control.py             # pick the config index when prompted
# cross-version experiments without editing configs:
ADDA_BINARY=../adda_1.4/src/seq/adda ADDA_VERSION=1.4.0 python adda_main_control.py
```

The config `local` branch maps `prefix→adda_python_scripts/`, `prefix_data→`repo root, so results appear under `./adda_results/…`. Keep probes small: one config with 1 T, 1–2 frequencies, 1–2 sizes (see

`adda_config_phase*_…py` for probe patterns). `force_new=True` forces recompute; otherwise completed jobs are skipped.

3.6 HPC runbook (Mistral / Thunder)

- Config hostname branch sets prefixes (`/work/um0878/…`, `/scratch/uni/u237/…`), `ramdrive=/dev/shm…`, Slurm `Partition`, `ncores`; submission goes through the site-specific `sbatch_simple*` wrappers (paths hard-coded in the main controls).
- Cluster configs still default to **ADDA 1.2 MPI** (`adda_1.2/src/mpi/adda_mpi` via `srun -n N`); the production decision is **1.4.0** — build an MPI 1.4 binary on the cluster and flip the triple, or export `ADDA_BINARY/ADDA_VERSION`.
- **Parallel job structure (Phase-4b recommendation, report §4b.3):** the per-vertex ADDA solves are process-level parallel and single-core-efficient — pool *N_cores* single-core `adda` invocations inside each (T,f,size) job rather than one MPI-parallel solve; keep the adaptive icosphere refinement sequential (it is a decision loop, not a loop nest).
- Cost anchors for planning: see §4.4 backlog table and `fig07`; the 1.4/1.5 trees also accept the old CLI subset unchanged (report §1.3 compatibility audit) except the `-orient avg` + grid-flag conflict (§1.5.5).

Part 4 — Validation & quality gates

4.1 Golden baselines (`baselines/`)

The frozen regression reference: 6 ADDA cases mirroring the exact CLI pattern of `acf.adda_call` (`-orient 0 0 0`, `-prop 1 0 0`, `-eps 3`, Matzler ice at 230 K): sphere (independent Mie check), oblate ellipsoids x=0.1/1/5, HongPlate_Id9 Dveq10 μm at 192/337 GHz. Current golden: `baselines/adda_1.4.0/` (regenerated 2026-11-06); `adda_1.2/` archived for provenance. Case results, Mie validation and scattering-pattern figures: **Appendix E**; narrative report: `baselines/REPORT.md`.

```
python baselines/make_baseline.py --adda <adda-seq-binary> --out baselines/<tag>
python baselines/check_baselines.py baselines/<tag>     # analytic sanity (sphere/Mie
↪ etc.)
```

The gate rule is noise-based, never bit-level. Measured ADDA re-run scatter of *identical commands* at `-eps 3`: 0.0 for four of six cases, $\leq 10^{-11}$ generally, but $8.3 \cdot 10^{-4}$ **for the** $x = 5$ **case** (12 312 dipoles at residual 10^{-3}). Therefore regression tolerances are: 10^{-10} for small cases, $5 \cdot 10^{-3}$ for $x = 5$. Any stack change (numpy/scipy bump, ADDA switch, symmetry fill-in) must land inside those bands. Do not regenerate goldens casually; a new golden requires re-deriving the noise budget.

4.2 Equivalence suites (what was proven, and how)

gate	command / artifact	result
Phase-2 numpy-2 migration	`analysis/phase2_equiv.py` (pre/post pickles)	pipeline bit-identical on numpy 2 / py3.12 / macOS
ADDA 1.2 vs 1.4.0 vs 1.5.0-alpha3	`analysis/phase3_equiv.py`, `phase3_results.json`	ext rel-diff $3 \cdot 10^{-17} \ldots 2.5 \cdot 10^{-4}$, mueller $\leq 5 \cdot 10^{-4}$ (all below the $5 \cdot 10^{-3}$ noise bound; x5 numbers *below 1.2's own scatter*); auto dipole density identical across versions; TRO `-orient avg` CrossSec **bit-identical** → **production binary: 1.4.0**
mirror machinery (sym vs asym, same config otherwise)	`analysis/phase4_subcontrol_gate.log`, `phase4_aoa_gate.log`	identical within $4.5 \cdot 10^{-12}$ with *closed* mirrors; a non-closed mirror (coarse YZ, closure 0.986) produces $1.5 \cdot 10^{-4}$ field errors — proof the closure gate is load-bearing
half-range β/θ reuse hypothesis	`phase4b_halfrange_check.py/.log`	**rejected**:$2.3 \cdot 10^{-3} \ldots 7.2 \cdot 10^{-2}$ discrepancy vs 10^{-11} noise
TRO β-fold (0–90° avg file vs full range)	`phase5_tro_beta_gate.py/.log`	CrossSec $\leq 1.35 \cdot 10^{-5}$ (= Romberg tol), Mueller $\leq 10^{-13}$ → PASS
ADDA cost/accuracy	`phase4b_adda_benchmark.py`, `…_bench_results.json`	fig. 04; error budget discretization-dominated
backscatter matrix (Phase 6)	`analysis/phase6_backscatter.py`, `phase6_results.json`	Mie pole agreement + convention pin + mirror-ring rule + SSDB backMat round-trip — all PASS

gate	command / artifact	result
Mueller artifact → NetCDF4 (Phase 7)	`analysis/mueller_nc_roundtrip_check.py` (20/20), `analysis/mueller_nc_gate.py` (W/X/R/E gates)	nc round-trip **bitwise** lossless; probe run + double-reader (`nc` vs legacy `pkl`) averaged dicts, phase chain and 5-file SSDB export (125 vars) **bit-identical**; 12/12 local `.pkl` artifacts migrated + bitwise-verified
analytic-grid twin (Phase 8)	`analysis/phase8_scatgrid_twin.py`, `phase8_endtoend_twin.py`, `phase8_export_twin.py`	adaptive vs static grid: all PASS (margin ≥4); full-pipeline 22×42 vs 181×361 twin: d_ext $7.6 \cdot 10^{-16}$, d_sph $1.9 \cdot 10^{-3}$; export cap keeps content bitwise, **15**× smaller beta files
forward-peak convention anchor (Phase 9A)	`analysis/phase9_fwdmodel_scan.py` (G1), `phase9_results.json`	Mie triple-anchor $\leq 5 \cdot 10^{-16}$; ADDA Cext vs analytic 0.23%/1.33% at dpl 70; calibration c=1.005/1.008 (cross-x dev 0.55%); far-field origin = scatterer center (0.000 lattice); port check $\leq 7.4 \cdot 10^{-11}$

Harness lessons that produced *false* failures — remember them when writing new gates: auto dpl is **quantized** (a 10^{-8} λ typo changes N_dipoles and breaks comparisons — compute λ/eq_rad exactly like `make_baseline.py`), and a scalar `theta=90.0` emits `z=6.12e-17` instead of exact `-prop 1 0 0`, rotating the φ reference (up to $2 \cdot 10^{-2}$ Mueller difference at x=5 — use the vector branch). Third trap (report §8.8/§9.2): *relative* per-element gate metrics explode on cancellation channels — Rayleigh-limit Mueller channels are pure noise; gate against the cell's physical intensity scale plus the store's measured noise plateau.

4.3 Where the accuracy lives (error budget recap)

$$\varepsilon_{\text{total}} \approx \underbrace{\varepsilon_{\text{disc}}}_{\text{fmin class}/dpl} + \underbrace{\varepsilon_{\text{orient}}}_{\delta=0.01,\ \geq 3 \text{ levels}} + \underbrace{\varepsilon_{\text{SH}}}_{\texttt{eps_sphexp}=0.005;\ \text{ph:}10^{-3}} + \underbrace{\varepsilon_{\text{solver}}}_{\texttt{-eps 3}:\ 10^{-11} \ldots 8 \cdot 10^{-4}} + \underbrace{\varepsilon_{\text{fp}}}_{\text{float32/c64}}$$

with the **discretization term dominant** (design margin $\geq 10^{-3}$).

If a consumer complains about smoothness/positivity (negative absorption vector a1), look at the SH truncation terms and the rescale diagnostics (§3.3), not at ADDA.

4.4 Production backlog (as of Phase 5)

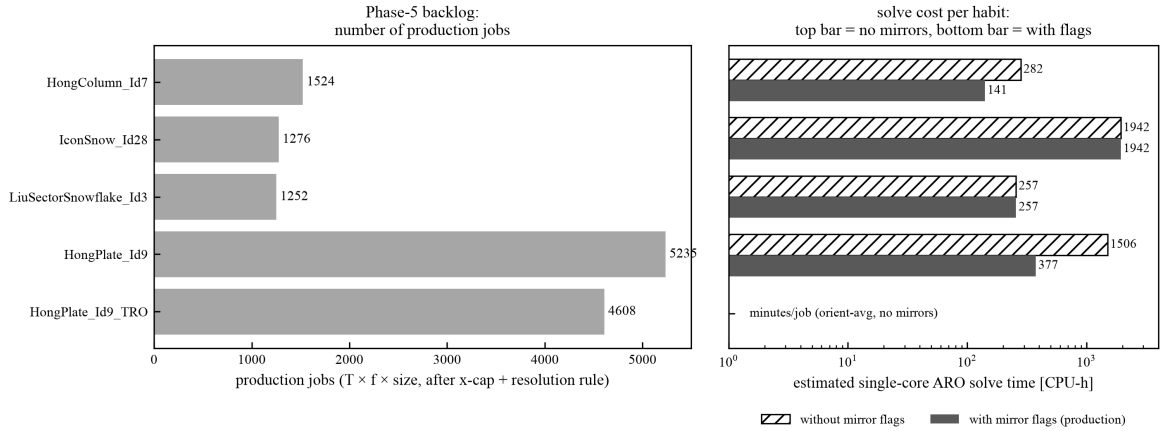


Fig. 7 — size and cost of the remaining production backlog. *Left: jobs per habit (each job = one (T, f, size), already after the x-cap and the resolution rule of Figs. 2–3). Right: estimated single-core solve time — the gap between the hatched bar (no mirror flags) and the solid bar (production flags) is what the mirror flags (§1.7) save; habits without usable mirrors (Icon/Liu) show no gap. TRO is cheap by construction (`-orient avg` inside ADDA).*

config	jobs	mirrors	est. single-core solve-time
HongPlate_Id9 (ARO, full 35-ch grid)	5235	XY+XZ (25 % solves)	377 CPU-h (unmirrored: 1506)
HongColumn_Id7	1524	YZ (50 %)	141 CPU-h (282)
IconSnow_Id28	1276	none	1942 CPU-h
LiuSectorSnowflake_Id3	1252	none	257 CPU-h
HongPlate_Id9_TRO	4608	(β-folded avg)	minutes per job

≈ **2700 single-core CPU-h** total. Runbook: local mode only for probes; production goes through the sbatch path with 1.4.0-MPI once built on the cluster (§3.6); ARO post-processing/export per §3.3. Inventory reproducibility: `analysis/phase5_backlog_inventory.py` (prints the table above and re-derives the job list with the exact control-selection logic).

Known open items (from report Phase 5, §5.7)

- Cluster MPI build of ADDA 1.4 (until then cluster runs use 1.2 defaults).
- TRO SSDB export path (currently `-orient avg` output is raw ADDA zips; a parser → NetCDF step is still to be specified).
- `sym_Zobj` parked (§1.7); half-range reuse rejected; GL-node β/θ grids rejected (no solves saved) — do not reopen without new evidence.

Appendix A — ADDA command-line tokens used

Full assembled command lines (verified from `acf.adda_call` / the TRO pair):

ARO solve (per icosphere vertex):

```
[launcher] <adda_binary> -shape read <file.adda> \
    -store_scat_grid -scat_grid_inp <scat_params.dat> -scat_matr both \
    -m <Re(m)> <Im(m)>  -lambda <μm>  -eq_rad <μm, equivalent-volume RADIUS> \
    -orient <alpha0> <beta0> <gamma0>   -prop <vx vy vz (icosphere vertex unit vector)> \
    -dir <tmpdir>/temp  -eps <3>
```

TRO solve (per (f, size, T) — orientation averaging inside ADDA):

```
[launcher] <adda_binary> -shape read <file.adda> \
    -scat_matr both  -orient avg <avg_params.dat> \
    -m … -lambda … -eq_rad …  -dir <job folder>  -eps <3>
```

token	note
`-eq_rad`	equivalent-volume radius (μm) — `Sizes` holds *diameter*, the sub-control halves it; N_dipoles comes from the shape file, ADDA only rescales
`-orient alpha beta gamma`	zyz Euler degrees (production 0 0 0); the *beam* direction comes from `-prop`
`-prop`	incidence unit vector; **always pass the vector branch** (a scalar 90° emits z=6.12 · 10^{-17} and rotates the φ reference)
`-store_scat_grid -scat_grid_inp`	181 × 361° grid from `scat_params.dat`
`-scat_matr both`	amplitude + Mueller on the grid
`-eps 3`	residual 10^{-3}; noise floor §1.1
removed from configs	`-Csca`, `-alldir_inp` (never parsed; 1.4/1.5 flag-combination traps)
version caveat	`-orient avg` cannot combine with `-scat_grid`/`-Csca`/`-alldir` in 1.4/1.5

Parsing contract: `ampl_scatgrid` 1 header line, forward row first, column order s22, s11, s12, s21 (pairs real/imag); `mueller_scatgrid` 1 header line, cols θ, φ, then 16 elements row-major — **all positional**, re-verify on any ADDA upgrade. The Phase-6 backscatter slice needs **no extra CLI token**: it is the Θ=180° row × Φ=0° column (plus the full Φ ring) of `mueller_scatgrid`; `acf.adda_call` raises if the scat grid (Appendix B) lacks a Θ=180° row or Φ=0° column — the production 181×361 grid has both.

Appendix B — the .dat input files

scat_params.dat (adda_python_scripts/, handed via -scat_grid_inp): global_type=grid, θ range N=181 (0–180°), φ range N=361 (0–360°) — 1° resolution; the phi_integr section belongs to ADDA's internal integration (unused by us since -Csca was removed).

avg_params.dat (adda_aux_input/<habit>/, TRO): per-Euler-angle Romberg spec min/max/Jmin/Jmax/eps/equiv/periodic (zyz). Production file for the TOR grid folds β **to 0–90**° using the XY-mirror closure (equivalence-gated $\leq 1.35 \cdot 10^{-5}$, §1.5.5); the full-range twin avg_params_fullrange.dat exists as the gate reference. Rule of thumb from the file comments: axysymmetrical particles → gamma max=0; equivalence planes → fold the range.

Appendix C — SSDB NetCDF layout

Filename: `Freq{f:08.3f}GHz_T{T:03.0f}K.nc`, one per (β, size, f, T), under `SSD/<orientation_type>/<medium>/<crystal_type>/<pristine_type>/<habit>/AzimuthallyRandom_beta{BETA:03.0f}deg/<Dmax…Dveq…Mass…>/`; root attrs `date`, `version='beta_0.9.1'` (Phase-6 additive backscatter bump; the schema below is otherwise the `beta_0.9` format of Brath et al. 2020); groups `SingleScatteringData`, `ShapeData`, `CalculationData`.

dimension	length	meaning
`sph_coeffs_scat`	common SH layout size	number of shtns coefficients of the phase-matrix re-expansion
`aa_inc`	1	azimuth of incidence (fixed 0°)
`za_inc`	19	zenith incidence-angle axis (°)
`phaMatElem` / `extMatElem` / `absVecElem`	16 / 3 / 2	element selectors (ext: K11, K12, K33; abs: 2 elems)
`scatMat_row`/`scatMat_Col`	4 / 4	Stokes matrix indices

`SingleScatteringData` variables: `frequency` (Hz), `temperature` (K), `za_inc`, `aa_inc`, index maps, **`phaMat_data_real + phaMat_data_imag`** (float32, m^2, zlib; dims phaMatElem ×aa_inc ×za_inc ×sph_coeffs_scat), `extMat_data` (m^2), `absVec_data` (m^2), **`backMat_data`** (float32, m²/sr, dims phaMatElem × aa_inc × za_inc) and **`backMat_index`** (int8, 4×4 fixed identity pairing) — the Phase-6 exact-backscatter matrix, stored independently of the phase-matrix spectra (§1.4 item 3).

`ShapeData` (0-d): `diameter_max`, `diameter_vol_eq`, `diameter_area_eq_aerodynamical`, `mass`, `dpl`, `N_dipoles`, `refrIndex_real/imag`, `alpha`, `beta`, `gamma` + attrs (`shape_file`, `refrIndex_model='Maetzler2006'`, `habit_id`, `medium`, …).

`CalculationData` (attrs): `method='DDA'`, `software='ADDA'`, `software_version`, `system`, `n_cores`, `date_completion`, `ADDA_eps`, `ADDA_scatParam_file`, `ADDA_avgParam_file`, `Refinement_delta`, `Maximum allowed refinement level`, `Mueller_sph_eps`, `Phase_sph_eps`, `sym_XYobj/XZobj/YZobj`, shtns version string. Missing config keys degrade to `'n/a'` (`getattr(ac, …, 'n/a')`) — never crash the export.

Appendix D — Gotcha catalog

Numbered for reference; several carry the bug-fix that neutralized them.

1. **Linux ELF binaries in `adda_1.2/src/seq|mpi`** — cluster-built, run nowhere locally. Always rebuild (`make seq`, NO_FORTRAN + FFTW rpath, §3.5); `make cleanseq` first or stale Linux `.d` files abort the build.
2. **Output parsing is positional** (`ampl_scatgrid` col order s22,s11,s12,s21!) — re-verify on ADDA upgrades.
3. **ADDA reruns are not bit-reproducible** at `-eps 3` (up to $8.3 \cdot 10^{-4}$ at x=5) → all gates noise-based.
4. **`-orient avg` + `-scat_grid`/`-Csca`/`-alldir` rejected** by ADDA 1.4/1.5.
5. **typhon must stay 0.10.0**; `ty.arts.xml` is gone — internal Mueller artifacts are **NetCDF4** (`*_mueller_data.nc`, schema in report §7.1; `rawdata2average` falls back to legacy `.pkl` with a deprecation warning); idempotency check accepts `.nc` plus legacy `.xml`/`.pkl` names.
6. **shtns on osx-arm64**: conda-forge has no build; rebuild wheel in-env (`--no-build-isolation --no-deps`) against the env's fftw.
7. **numpy-2 migration done** (Phase 2): `np.nan`, `np.complex128`, `trapezoid`, `np.tile`; `posix_spawn.py` has a macOS subprocess branch. Never reintroduce old aliases.
8. **Config globals are exec-injected** — renaming a config global breaks the drivers silently; keep the field contract (§3.1).
9. **Deprecated, do not extend**: `adda_sub_control_gridref.py`, `acf.romberg_*`, `adda_orientational_avg_spharmonics_for_testing.py`, `old_configs/`, `SSD_scripts_deprecated/`.
10. **TRO `liste` must have 9 rows** (row 8 = temperature). With 8 rows the folder name gets the flat index as `T####K` — now guarded by a loud warning; rebuild old-style configs instead.
11. **TRO has no idempotency**: chunk ranges are re-run unconditionally; TRO `deq` folder field is the eq **radius** (legacy).
12. **Mirror flags from the coarsest class only**; closure <1.000 fill-in = real shape error (~$1.5 \cdot 10^{-4}$), not noise. Re-run `analysis/phase4_symmetry.py` when adding or regenerating shape files.
13. **f < 192 GHz in the frequency list runs** (coarsest class, grossly over-resolved) — the `=np.array([5,])*1e9` stub configs are pointless work, and `analysis/inventory.py`'s per-file rule counts them as zero (mismatch with the live control's behavior at low f — report §1.7 wording refers to the replica).
14. **`-prop` scalar vs vector branch**: always pass the unit vector (scalar 90° ⇒ z=$6.12 \cdot 10^{-17}$ ⇒ φ-reference rotation ⇒ $2 \cdot 10^{-2}$ Mueller artifacts in comparisons).
15. **`export2SSDBnetCDF` is off by default** (`SSDB_EXPORT_NETCDF=1`) and habit-selected by `SSDB_HABIT`; the 19 × 19 grid + `eps_phase=0.001` are hard-coded in `adda_process_aro_particles.py`.
16. **Local mode is synchronous** — `adda_main_control.py` on the Mac runs jobs one-by-one in the foreground; use probe configs, and pool single-core ADDA solves if you parallelize (§3.6).
17. **Per-size class availability is not guaranteed complete** (e.g. HexPlaAgg 197 μm lacks the 192/337/457 classes); and for small particles the five classes are degenerate copies (identical N_dipoles).
18. **Vertex-cache is transient**: `<shape>_temp_data_vertex_*.nc` deleted at job completion; killing a job mid-way leaves it (useful), completing wipes it (startover of a *single vertex* re-solves).
19. **ZIPPING=False** silently deletes the auto-created vertex zip (ADDA raw output is not kept); set `ZIPPING=True` when debugging a solve.
20. **No version control was present historically** — repo is a git checkout now; still: results/intermediates live outside the repo (`adda_results/…`), and old HPC results are unrecoverable.
21. **Phase-6 backscatter data**: the raw `back_matrix`/`back_mat_ring` are **k-normalized** (dimensionless) — never treat them as µm²/sr before the `rawdata2average` $\lambda^2/4\pi^2$ conversion. Vertex caches / overview mats predating Phase 6 carry no back keys → treated as cache misses (forced ADDA re-

runs); `rawdata2average` on old stores warns and leaves `backMat` NaN. The B-matrix polarization convention is $R(+\varphi)$ on **both** sides — using the extinction branch's $-\varphi$ is wrong (gate floor 3e-2 vs 0.96 error).

22. **Near-pole phase-matrix off-diagonals carry a few-% pole-interpolation artifact** (pipeline `mueller2phase`, known): if a consumer cross-checks `backMat` vs phaMat($180^\circ - \theta_{\mathrm{inc}}$), expect that floor — `backMat` at the exact point is the cleaner value (report §6.5).
23. **netCDF4 raises `IndexError` (not `KeyError`) for missing variables** — the cache readers catch both; a pre-Phase-6 store previously crashed `read_data_from_netcdf` (caught by the Phase-6 gate).
24. **Mie reference conventions (Phase 9A, pinned):** the B&H 5.12 log-derivative (D) form with *downward* recurrence (as in `analysis/phase6_backscatter.py`) is **validated** (== 40-digit mpmath == miepython == published Wiscombe MIEV0 case-14). Do **not** "fix" it with an explicit psi/chi/tau series replacing $D_n(mx)$ by τ_n'/τ_n — that is not a valid B&H formula (gives negative Qext for lossy m; an earlier debug round chased it). Angular series: $\pi_n = -\mathrm{lpmv}(1, n, \mu)/\sin\theta$ (scipy lpmv carries the CS phase), $\tau_n = n(n+1)P_n(\mu) - \mu\pi_n$, pole limits $\pi_n(\pm 1) = \pm n(n+1)/2$ signed.
25. **own-eps SH ntrunc is misleading for hybrid/residual products** (relative tolerance on a small field): judge residuals at the FULL-field M11 scale (report §9.2). Log-space convention fits are poisoned by model zeros — fit in amplitude space.
26. **ADDA far-field phase origin = scatterer center** (fitted lattice offset 0.000, Phase 9A G1) — analytic-model ports must center on the scatterer, not the voxel-box corner (the `mueller_scatgrid`-vs-ampl port check at ≤7e-11 confirms the rest of the frame port).
27. **Size-grid shape globs are 4-digit zero-padded** (`Dveq01078um`, not `Dveq1078um`) — a bare-number glob silently matches nothing (or worse, the wrong class).
28. **At $x \gtrsim 10$ the full-field SH content saturates the static grid's Nyquist cap** (ntrunc 179 ≈ 180 on the 361-φ ring): large-x full-field twins are grid-limited, not expansion-limited (report §9.3).
29. **History rewriting physically DELETES the purged paths from the working tree.** `git filter-repo` ends with `git reset --hard` on a non-bare repo (its `cleanup()` sets `reset = not GitUtils.is_repository_bare(...)`), so a path it removes from history is erased from disk — **gitignore does not protect it**, because gitignore never applies to entries in the index. Hit on 2026-11-10: purging `adda_shape_files/` wiped the 1.8 GB live copy (recovered from the pre-purge bundle, 1940/1940 files hash-verified — report §10.4). Rule: untrack first (`git rm -r --cached <path>` + commit) or rewrite in a **bare** clone, and always `git bundle create ../backup.bundle --all` first.
30. **`adda_shape_files/` is not in git** (1.8 GB of external input, purged 2026-11-10, report §10): a fresh clone cannot run any habit config, gate suite or baseline until the folder is provisioned by hand (§1.2 "Shape-data provisioning"; restore recipe also in `agents.md`). Never `git add -f` it back — that reintroduces the 170 MB blob and the unpushable history.
31. **The ADDA trees are not in git either** (untracked + purged 2026-11-10, report §11): `adda_1.4/` (1.4.0 = **production** binary, upstream tarball), `adda_1.2/` (upstream v1.2 tarball, golden-baseline lineage), `adda/` (1.5.0-alpha3 = clone of `m-brath/adda`, upstream has no such tag; validated state `8f550a7`), `adda-gui/` (`m-brath/adda-gui`). A fresh clone therefore has **no solver** — fetch + build per §3.5 before any solve, gate or baseline. Configs locate the binary via `adda_prefix + adda_binary` (env override `ADDA_BINARY`), so the folder names above are contractual. The once-documented hard-coded `…/adda/adda_1.2/src/mpi/adda_mpi` in the sub-control script is obsolete (Phase 3 made the binary config-driven) — the hazard is the missing tree, not a wrong path.
32. **A gitlink without a `.gitmodules` entry is broken on GitHub and fatal to tooling.** A nested repo tracked as mode 160000 with no mapping shows up on GitHub as a folder that 404s (the link falls back to this repo's own path), and `git submodule` / `git clone --recursive` abort with "No url found for submodule path …". Fixed 2026-11-10: `SSD_scripts_deprecated` had no remote at all (its pinned commit was unresolvable anywhere) and was absorbed into the umbrella as plain files; the pipeline gitlink was removed entirely the same day (§33). Rules: push a submodule's commits to

its own remote **before** pinning it; never introduce a gitlink without its `.gitmodules` entry; to convert a gitlink into files, `git rm --cached <path>` first (plain `git add <path>` silently does nothing).

33. **The pipeline lives in this repo — do not re-nest it.** `adda_python_scripts/` (and `SSD_scripts_deprecated/`) are ordinary tracked files since 2026-11-10; see #34 — the repo has no submodules at all. The pipeline's full history (10 commits, 2017-06-13 → Phase 8) was grafted in: bundle → scratch clone → `git filter-repo --path-rename :adda_python_scripts/` → `git merge --allow-unrelated-histories`; verified by a blob-SHA tree diff against the pre-absorption tree `1152d0c2` (NONE, 87/87 identical). Archive of the original nested repo: `../adda_python_scripts-history-2026-11-10.bundle`. Consequence: pipeline edits are now commits in the DDA repo — no second repo to sync per phase (report §13).
34. **No submodules — do not add any back.** `arts-sspdb-interface/` was de-registered on 2026-11-10 (report §14): it keeps its own `.git`, its own `origin` (`m-brath/arts-sspdb-interface`, upstream `atmtools/arts-sspdb-interface`) and its own HEAD (`e7b71023` at untracking), and is gitignored exactly like the ADDA trees. The umbrella therefore has **zero gitlinks and no `.gitmodules`**: everything third-party is an independent checkout, everything project-owned is tracked here. If the tooling must ever become plain files, §13.2 of the report is the recipe.

Appendix E — Baseline simulation results

The frozen regression set behind §4.1: six ADDA runs mirroring the exact CLI of `acf.adda_call` (`-orient 0 0 0`, `-prop 1 0 0`, `-eps 3`, Matzler-2006 ice) — three analytic/analytic-voxel sphere & spheroid cases at x = 0.31/0.1/1/5 plus two HongPlate_Id9 (D_veq = 10 μm) production-pattern cases. Current golden: `baselines/adda_1.4.0/`; `baselines/adda_1.2/` is the archived provenance set — the figures below are rendered from the 1.2 outputs, but Phase-3 equivalence (§4.2) proves the 1.4.0 goldens agree within the noise bands. Full narrative: `baselines/REPORT.md`.

E.1 Headline numbers (frozen 1.2 outputs)

Cross sections from `CrossSec-X/Y`; ω = Csca/Cext with Csca from the (then-production) `-alldir_inp` Romberg integration — note today's ARO configs no longer request `-alldir_inp` (§1.4), these baselines predate that cleanup.

Case	x	N dipoles	dpl (λ)	Qext (X / Y)	ω (X / Y)
ice sphere, 300 GHz, 230 K	0.314	2 176	160.7	6.914e-3 / 6.914e-3	0.684 / 0.684
oblate spheroid c/a=0.5, 183 GHz, 253 K	0.1	1 104	402.8	3.824e-4 / 7.533e-4	0.0814 / 0.0820
oblate spheroid c/a=0.5, 183 GHz, 253 K	1	1 104	40.3	2.630e-1 / 6.612e-1	0.957 / **1.003**
oblate spheroid c/a=0.5, 183 GHz, 253 K	5	12 312	**18.0**	1.574 / 2.217	**1.569 / 2.091**
HongPlate_ID9, Dveq=10 μm, 192 GHz, 230 K	0.020	1 015	1947	6.516e-5 / 1.015e-4	9.19e-4 / 9.35e-4
HongPlate_ID9, Dveq=10 μm, 337 GHz, 230 K	0.035	1 015	1109	2.052e-4 / 3.195e-4	2.77e-3 / 2.82e-3

Consistency checks that pass: sphere Qext X ≡ Y to all printed digits (rotational symmetry); every habit case shows the expected X-vs-Y polarization splitting; the two Hong-plate runs differ only via m(f) and the Romberg grid, Qext scaling ≈ ×3.15 from 192→337 GHz (Rayleigh: Cext ∝ Im(m)·f — matches).

Sphere vs. independent Mie theory (Riccati-Bessel in `check_baselines.py`, cross-checked against `miepython`):

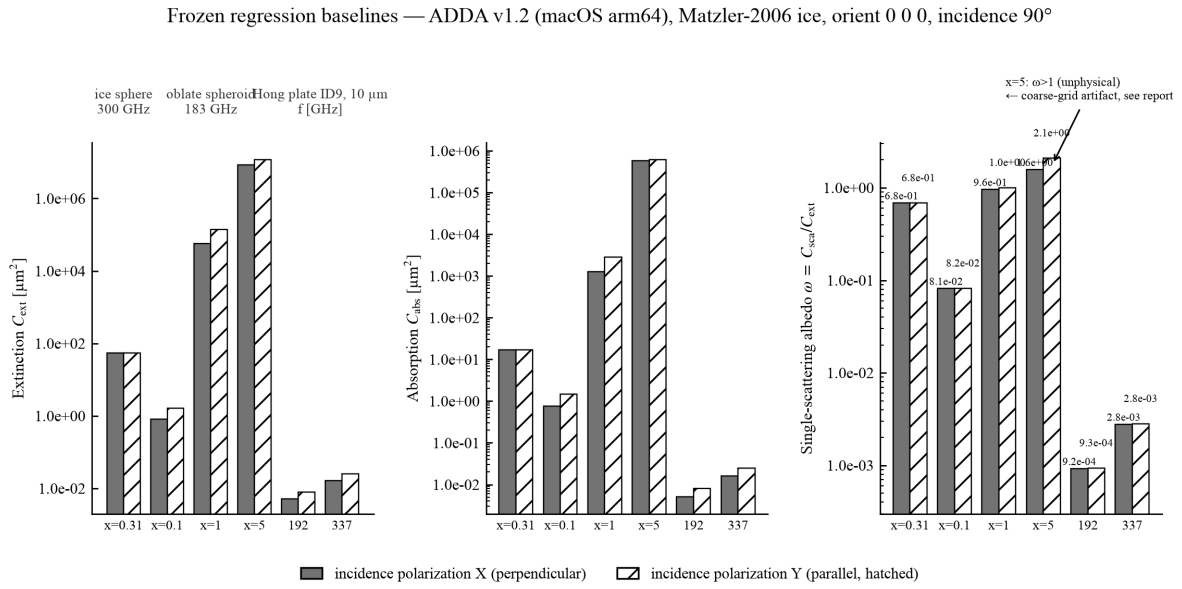
quantity	ADDA (auto grid)	Mie	deviation
Qext	6.9144e-3	6.7929e-3	+1.79 %
Qsca	4.7310e-3	4.6539e-3	+1.66 %
Qabs	2.1865e-3	2.1390e-3	+2.22 %

All inside the 3 % acceptance band, and the deviation is *confirmed DDA discretization error*: forcing `-dpl 320/800` converges Qext 6.914e-3 → 6.852e-3 → 6.819e-3 toward the Mie value (fig. E-2). Cross-version:

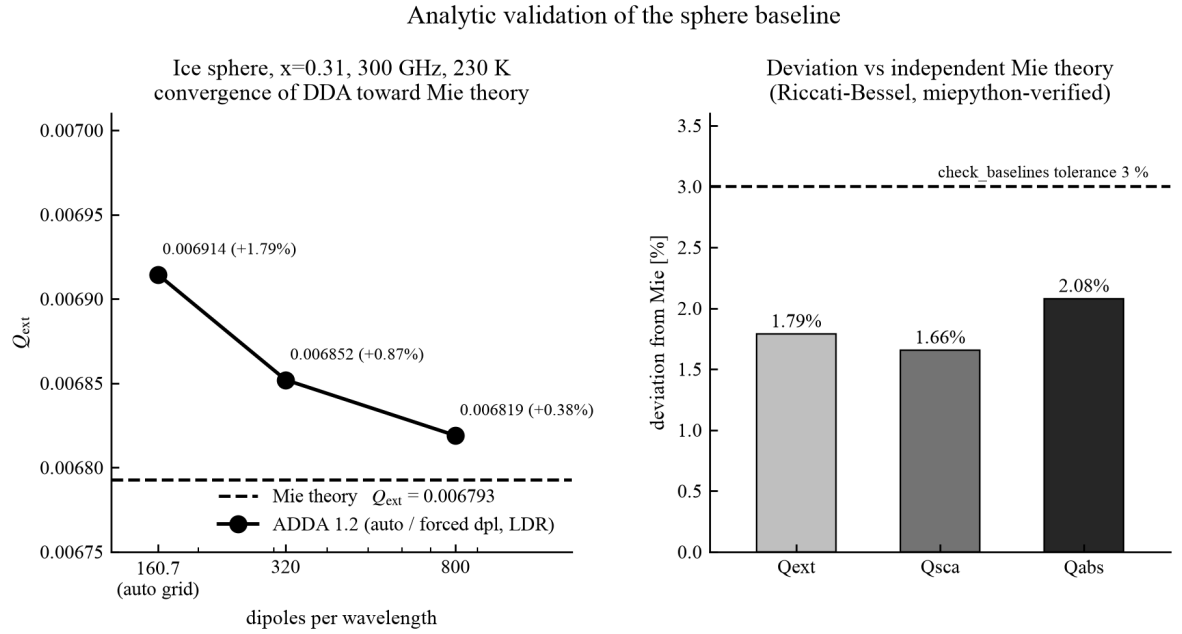
ADDA 1.2 vs 1.5.0-alpha3 identical Qext for the sphere to 12 significant digits; cross-platform: the macOS arm64 rebuild of 1.5.0-alpha3 reproduces the Linux cluster numbers exactly.

Caveats (report §4): ω > 1 at x = 5 is unphysical — Csca exceeds Cext by 68 % at the auto grid (18.0 dpl, outside LDR validity ≲6–10 dpl); the angular integration is fully converged, so this is a dipole-grid artifact. The case remains a valid deterministic *regression fingerprint*, but its absolute Csca/Qsca/ω are not physics references — the same caution applies to production runs at the large-x end of the backlog (§4.3 discretization-dominant budget). The x=1 ω=1.0025 value is the same effect at 0.2 % — keep it as a regression tripwire.

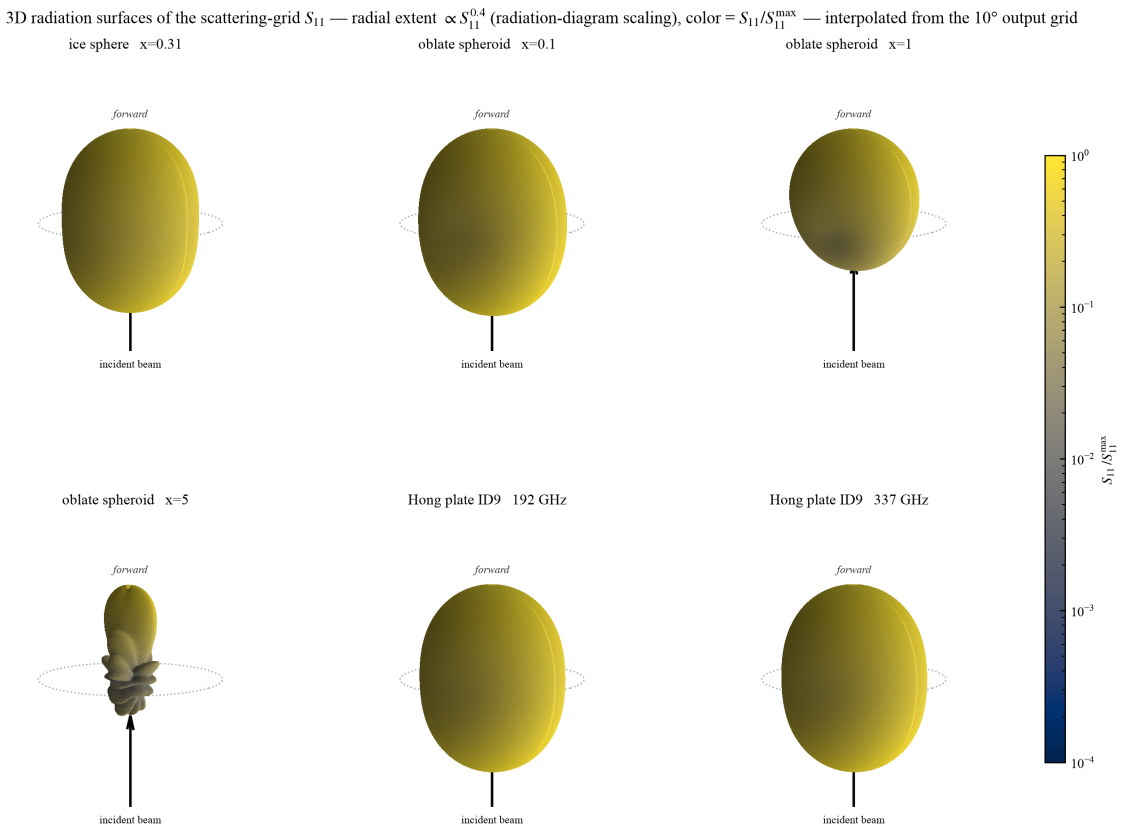
E.2 Figures



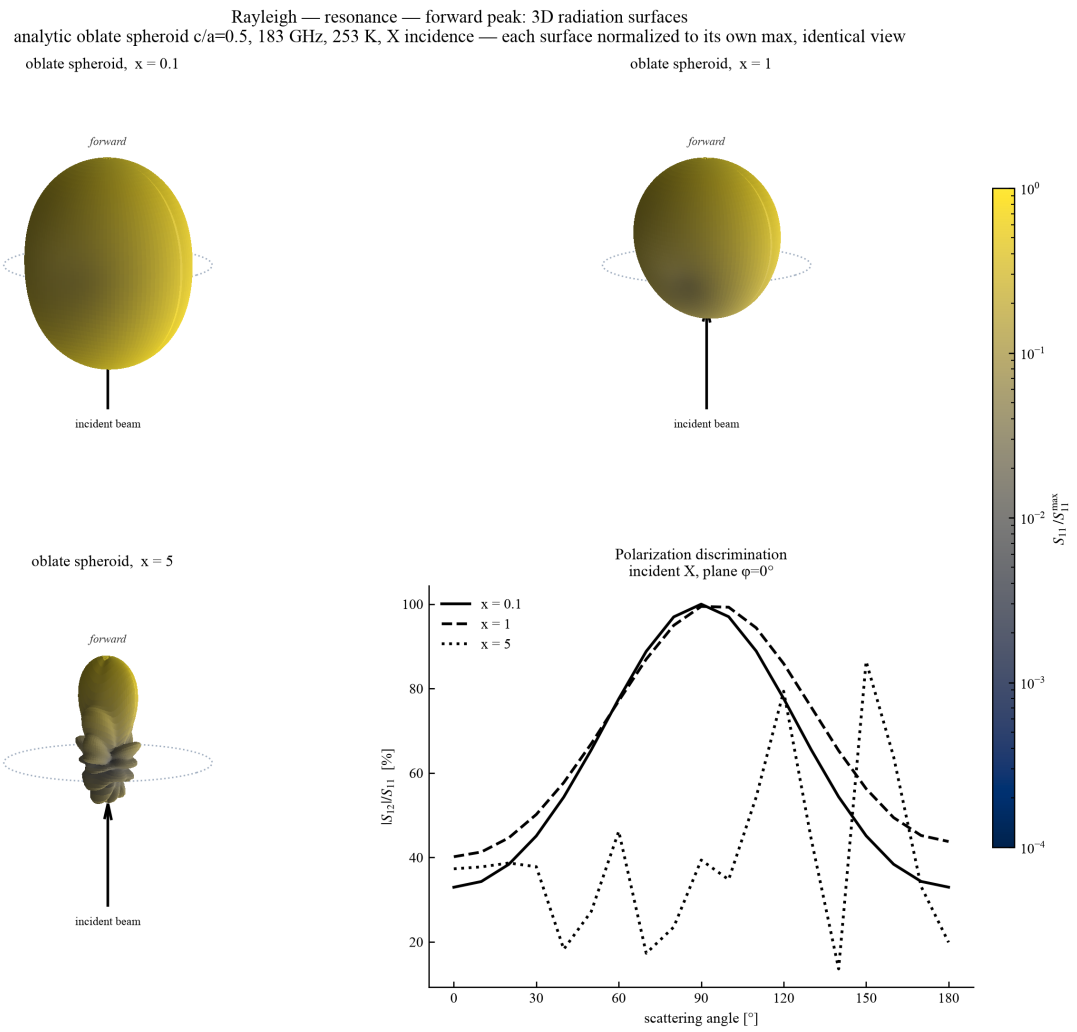
E-1. Cross sections & albedo, X vs Y incidence polarization, all six cases.



E-2. Analytic validation: the sphere baseline converges toward independent Mie theory with dpl.



E-3. 3-D radiation surfaces of the scattering-grid S_{11} (radial extent $\propto S_{11}^{0.4}$, color = S_{11}/max, beam axis marked; interpolated from the 10° output grid). The near-Rayleigh particles (sphere, x=0.1 spheroid, both Hong plates) are essentially isotropic shells; x=1 develops diffraction rings; x=5 shows the classic narrow forward spike with ringed side lobes.



E-4. Rayleigh → resonance → forward-peak evolution of the analytic oblate spheroid (each surface normalized to its own max), plus polarization discrimination $|S_{12}|/S_{11}$ in the φ=0 plane.

E.3 Reproduce

```
mamba activate DDA
python baselines/plot_baseline_figs.py      # figures (PNG 250 dpi + vector PDF,
 ↪  baselines/report/)
python baselines/check_baselines.py         # Mie consistency, exit code = pass/fail
python baselines/make_baseline.py --adda <binary> --out baselines/<tag>   # rerun cases
python docs/make_figures.py                 # mirrors the PNGs into docs/figures as
 ↪  fig09..fig12
```

References

- **Brath, M., Ekelund, R., Eriksson, P., Lemke, O., Buehler, S. A. (2020)**: Microwave and submillimeter wave scattering of oriented ice particles, *Atmos. Meas. Tech.* **13**, 2309–2333, https://doi.org/10.5194/amt-13-2309-2020. — **The main methodology article for everything ARO in this repo**: TRO/ARO definitions, zyz Euler angles, the ARO averaging approach (DDA solves + own orientation averaging over α, γ), the (β, θ_{inc}) output coordinates, and the RT motivation (ICI, GMI 166 GHz polarization). Their ARO database: zenodo 3463003.
- **Eriksson, P., Ekelund, R., Mendrok, J., Brath, M., Lemke, O., Buehler, S. A. (2018)**: A general database of hydrometeor single scattering properties at microwave and sub-millimetre wavelengths, *Earth Syst. Sci. Data* **10**, 1301–1326, https://doi.org/10.5194/essd-10-1301-2018. — The published ARTS SSPDB (TRO) paper: source of the habit-ID numbering and shape-file lineage (`*_Id<n>` folder names), the 35-channel frequency selection, the 190/230/270 K temperature set, the Mätzler-2006 refractive-index choice, and the SSDB NetCDF format the export targets (zenodo 1175572, tooling: `arts-sspdb-interface/`).
- Yurkin & Hoekstra 2011, *JQSRT* 112, 2234–2247 (ADDA); ADDA manual: `adda/doc/manual.pdf`, `adda_1.2/doc/manual.pdf`.
- **Bohren, C. F., Huffman, D. R. (1983)**: *Absorption and Scattering of Light by Small Particles*, Wiley. — cited as **BH** in this manual: source of the $s_1..s_4$ amplitude convention behind ADDA's `ampl_scatgrid` (§1.4, §1.9.1), the extinction paradox and diffraction-peak treatment (§1.9.2), and the validity conditions of the diffraction/eikonal approximations (§1.9.5).
- **van de Hulst, H. C. (1957)**: *Light Scattering by Small Particles*, Wiley. — classic derivation of the extinction paradox via Babinet's principle and of the exact internal-current representation underlying the Phase-9 model (§1.9.2, §1.9.5).
- **Wiscombe, W. J. (1979)**: Mie scattering as a function of particle size, in *Physical and Quantitative Aspects of Radiative Transfer: An Introduction to the Eleventh Symposium* (K. S. Shifrin, ed.), NASA SP-234. — partial-wave cutoff $n_{\max} = x + 4x^{1/3} + 2$ (§1.9.3); his MIEV0 reference cases anchor the Phase-9 Mie validation (§1.9.5).
- Mätzler 2006, dielectric function of ice (implemented in `acf.eps_ice_matzler06`).
- Mishchenko 2002 (*Light Scattering by Nonspherical Particles*) — extinction-matrix construction used in `acf.extinction_matrix`.
- ARTS Microwave SSP Database: https://doi.org/10.5281/zenodo.1175572 (browse/import tooling: `arts-sspdb-interface/`).
- Project working documents: `plans/DDA-modernization-plan.md` (approved plan), `plans/DDA-modernization-report.md` (findings per phase — **read before code changes**), `baselines/README.md`.

Generated with the repo state of 2026-11-09 (figures redesigned for readability the same day). Figures regenerate via `docs/make_figures.py` (Appendix-E baseline plots delegate to `baselines/plot_baseline_figs.py`); this file plus `docs/DDA-manual.html` are produced from the same source.